

Solution Selection Form**Approved at February 15, 2024 SDC**[Link to Meeting Minutes](#)

Project Title: Hyde Park BESS – Battery Energy Storage System Solution	Project Number: 24211 (D Line), 21334 (D SS)
Date Prepared: 12/22/2023	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: D Line D SS
Project Initiator: Mariam Chatha	Project Category: Stations - Batteries
Managing Organization: Project Mgt and Construction	Schedule ID#: 80089926
Project Manager: Adam Kelley	Project Type: Specific
Project Sponsor: Lavelle Freeman	Project Purpose: A new Battery Storage System to Alleviate Area Loading Issue
Estimated in service date: September 2027	EJ Community Impact: Yes
Conceptual Grade Cost Estimate (preferred solution): \$49.2M	
Facility Type (check all that apply): ☐ PTF ☐ non-PTF ☐ local PTF ☒ Distribution ☐ N/A (General Plant) ☐ BPS ☐ BES ☐ CIP	

[1] Project Need Statement

The 13.8kV Hyde Park Substation is forecasted to have a steady increase in electrification load. The system needs for the northern area can be classified into several broad categories:

- **Capacity Deficiency:** Under an N-1 contingency of a Hyde Park 115/13.8kV transformer, the remaining Hyde Park Substation transformer is overloaded. Load is typically transferred during a Substation transformer contingency via distribution circuit ties utilizing automatic field switches. However, with this load transfer, the Hyde Park Substation load is still forecasted to contribute to overloads under an N-1 contingency by the summer of 2024.

Hyde Park Substation 2021 peak load exceeded the Substation firm capacity and almost exceeded the Substation Load Carrying Capability (LCC). The firm capacity is the capacity based on the loss of a Hyde Park Substation transformer and the Load Carrying Capability is the firm capacity in addition to the amount of load that can be transferred to nearby Substations utilizing distribution circuit ties. In 2021 and 2022, load was transferred permanently off of Hyde Park Substation after the Substation peaked in those years. However, even with these load transfers, the Hyde Park Substation is forecasted to exceed the firm capacity of the Substation and reach 95% of the Hyde Park Substation LCC by 2024.

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Circuit 496-H1XY runs 80-90% of its summer normal ampacity and circuit 362-1209H runs 60-90% of its summer normal ampacity in the summer months. Circuit 496-H1XY supplies Milton and Dorchester while circuit 362-1209 supplies Milton.

This need is described in more detail later in this document following an overview of the transmission and distribution system serving this area.

By July of 2027, the capacity to be delivered by the project 21334 will need to be delivered by other means. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours during the summer timeframe typically from 18:00 PM to 20:00 PM. Based on the 2023 forecast, two (2) generators of one MW capacity would be needed in 2027 and 2028 and two (2) generators of two MW capacity would be needed in 2029 until the BESS is installed. Eight MW of generation would be needed in the worst-case scenario where the BESS does not get installed until the station is at 100% of LCC. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time. System Planning will monitor the annual forecast that may shift the plan laid out above.

- **Transmission Reliability:** Various N-1 transmission contingencies may result in loss of the 115kV line supplying the Hyde Park 115/13.8kV transformer including: faults on the 115kV Baker St - Hyde Park lines; or 115kV bus faults at Hyde Park or Baker St; or 115kV breaker failures at Hyde Park or Baker St. These would cause the loss of a Hyde Park 115/13.8kV transformer, which would leave Hyde Park Substation supplied by only one remaining transformer. This results in capacity issues explained in the section above.
- **Distribution Reliability:** Circuits 496-H1XY and 362-1209H run up to 90% of their summer normal ampacity. This thermal constraint will be explained further in sections below.
- **Asset Condition:** Both the 110A and 110B transformers at Hyde Park are over 50 years old and are already in a 5 year long term plan for replacement.

Project Background

The Substations surrounding Hyde Park Substation are also heavily loaded including Dewar Street and Andrew Square Substations. Dewar St substation is forecasted to be above 100% of the firm capacity from 2024 onwards and Andrew Square Substation is forecasted to be above 100% of firm capacity by 2027. Additional capacity is needed in the area to resolve the Hyde Park Substation #496 overload and nearby Substations' heavily loaded conditions. A new Substation is considered as the long-term solution to address the area capacity issues. Currently the engineering team is focusing on evaluation of two (2) properties for the project. One (1) is the property on Cummings Highway in Hyde Park; the other property is at 161-191 New England Avenue in Boston, MA. These two locations are being considered for the future substation.

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The project 21192 – Hyde Park-Dorchester Area Supply Initiatives, which addresses the 10-year long-term solution was a partial funding PAF approved by EPAC on August 22nd. The PAF was revised to include the 15-year plan and was submitted to EPAC on October 24, 2023.

Below is a diagram of the two alternatives to relieve Hyde Park and surrounding Substations.

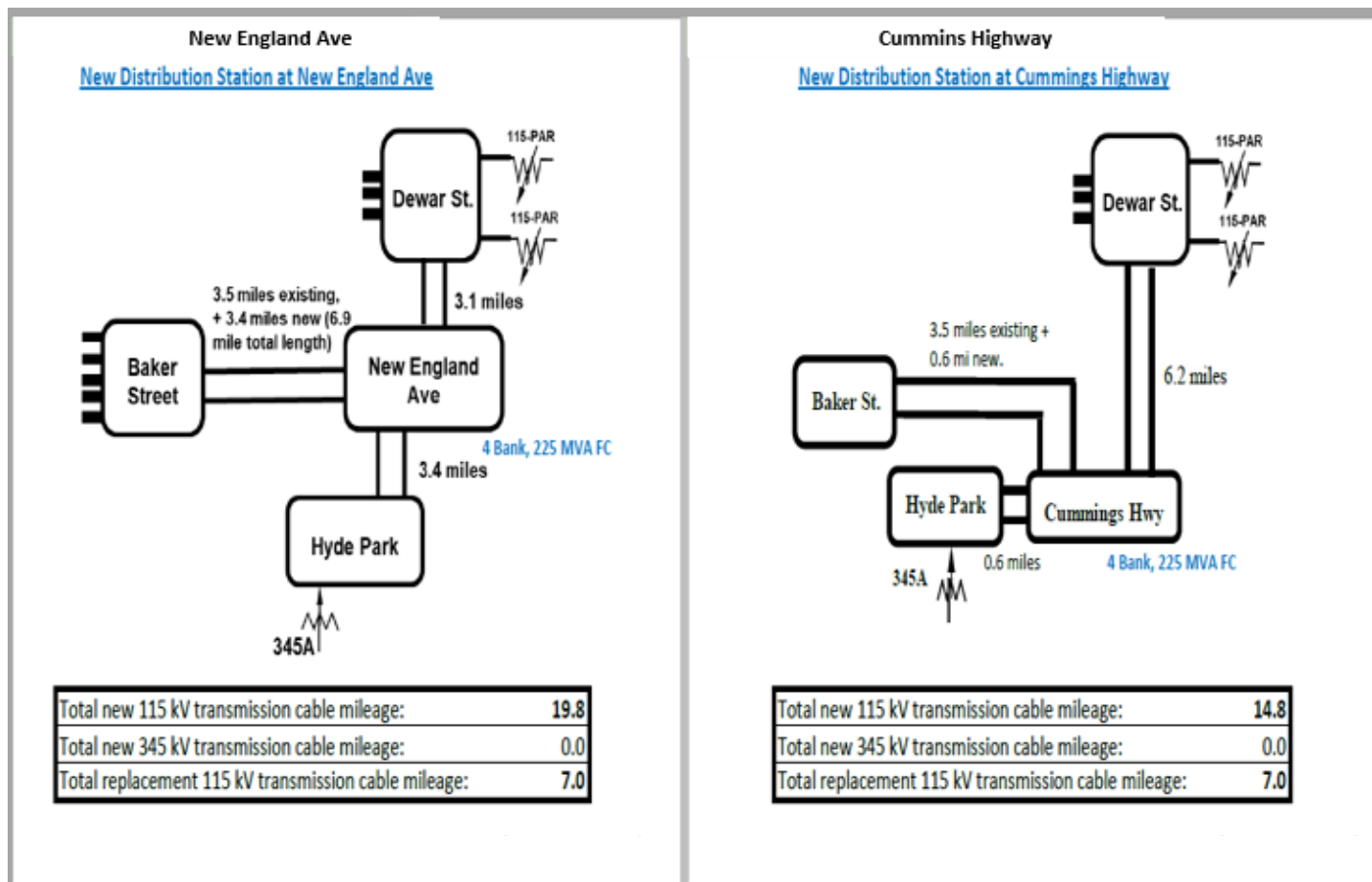


Figure 1: Alternatives for Hyde Park-Dorchester Area Supply Initiatives project

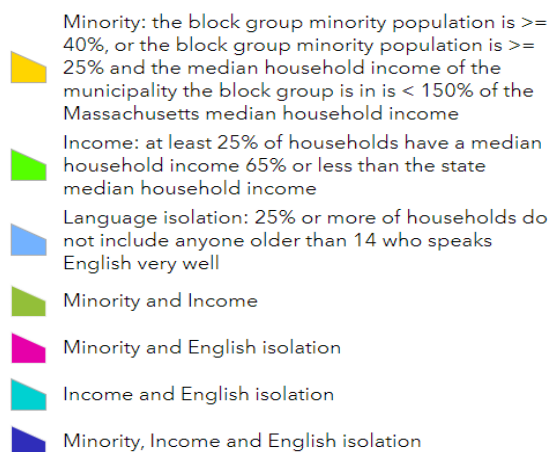
The project 22345 – Hyde Park-Dorchester Area Plan: Land Acquisition has purchased the property at 161-191 New England Avenue, Boston to address the capacity constraint in the area as a long-term solution. This project was approved by EPAC on September 12, 2023 and the property closed on October 24, 2023.

Building a new Substation to address the thermal constraint will take until December 2031 and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal Substation constraint until the long-term plan is in service, which is why this project is proposed.

Customer Overview

Hyde Park is the southernmost neighborhood of Boston, Massachusetts and is situated 7.9 miles south of downtown Boston. It is an urban location with suburban characteristics. Hyde Park Substation supplies 61,740 customers mainly in Hyde Park West Roxbury, Dedham and Milton as well as Brookline, Dorchester and Westwood. West Roxbury, Dedham and Milton are also towns in Boston, Massachusetts located adjacent to Hyde Park. The area is supplied by twenty-four 13.8kV feeders from a 345/115/13.8kV Hyde Park Substation.

Based on the Climate and Economic Justice Screening Tool (CJEST), Hyde Park and surrounding downtown Boston parcels include areas where the majority of the population is a minority and speak English as a second language. Please see figure 2 for the CJEST map for the Hyde Park supply area.



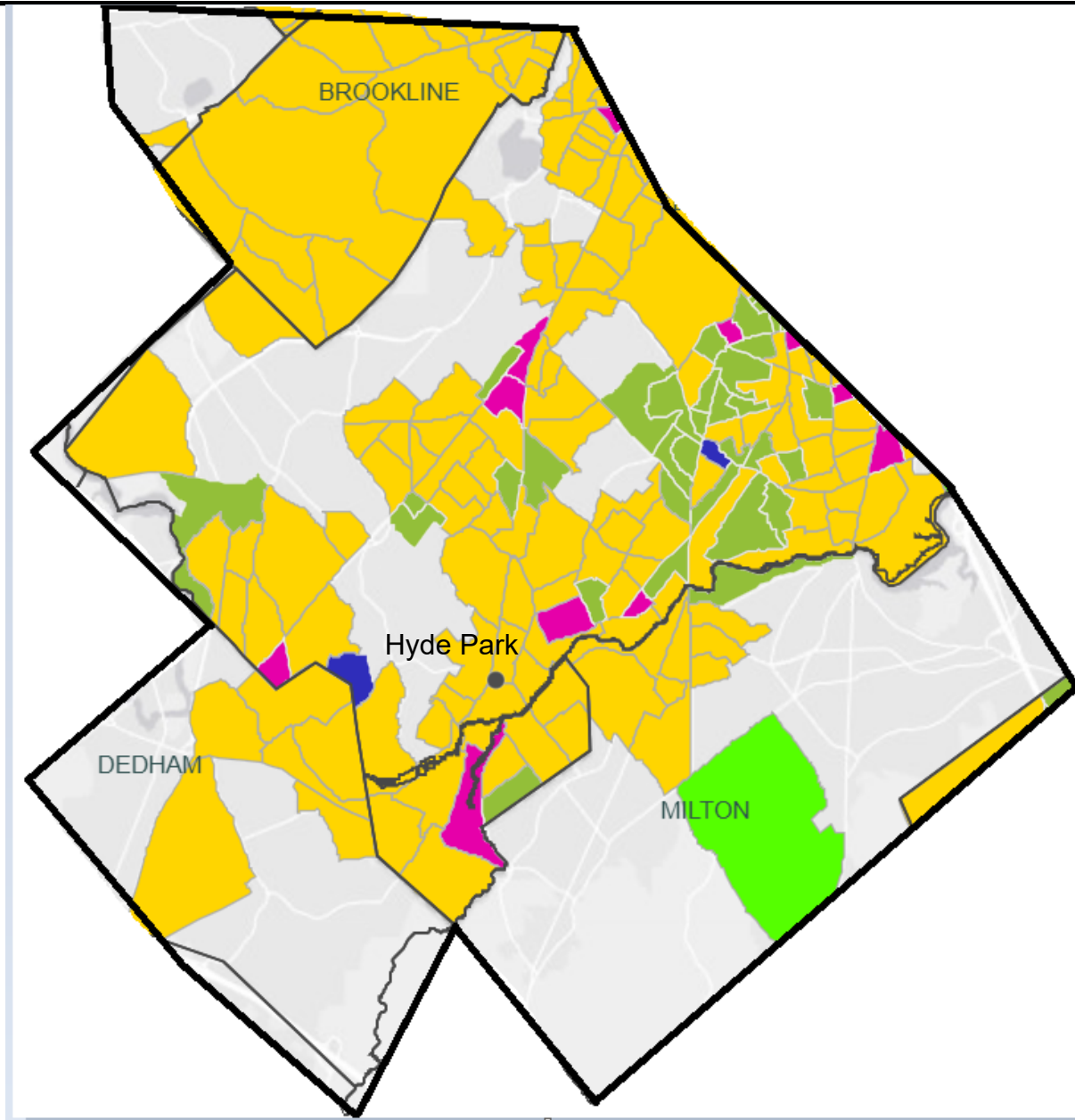
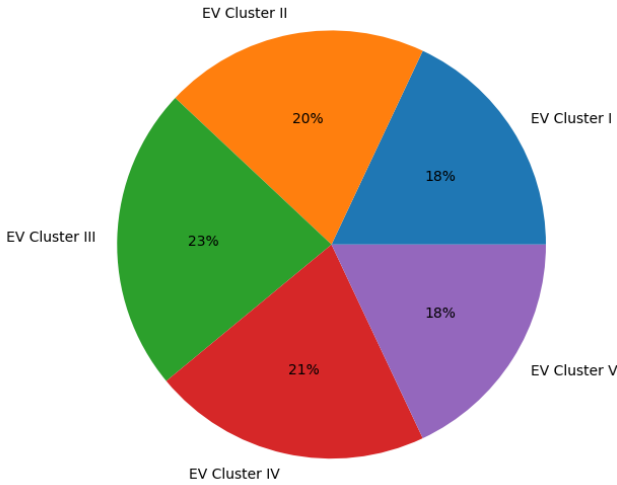


Figure 2 CJEST Hyde Park supply Area

EV Adoption Clusters for Station 496 Hyde Park



Heating Adoption Clusters for Station 496 Hyde Park

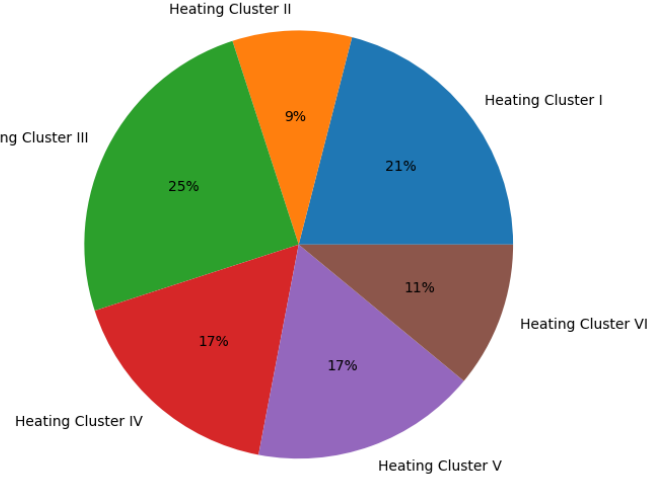


Figure 3 . The percentage of EV and Heating adoption clusters for Station 496 Hyde Park

Circuit	Heating Cluster I (in %)	Heating Cluster II (in %)	Heating Cluster III (in %)	Heating Cluster IV (in %)	Heating Cluster V (in %)	Heating Cluster VI (in %)
344-1411H2	0	0	10	14.3	34.3	41.4
441-1401H1	0	0	67.4	15.5	10.9	6.2
496-H3	12.4	8.4	22.7	27.9	12.6	16
496-H5	26.8	11.2	23.2	11.8	15.3	11.8
496-H7	33.5	11.5	25	9.5	12.2	8.3
60-1409J1	16.1	5.9	19.9	26.9	11.8	19.4
344-1411H1	0	75	0	25	0	0
496-H4	11.3	3.4	28.9	28.8	11.9	15.5
496-H8	16.8	13.3	16.6	29	12.1	12.2
344-1412H1	20.8	7.9	28.8	2.7	37.5	2.5
362-1207H1	38	10.6	33	0.7	16.5	1.3
362-1209H1	28.4	16.4	28.3	2.7	22.8	1.4
496-H6	22.5	8.8	30.2	12.5	16.1	10
496-H1XY	18.4	8.6	30.3	20.2	12.2	10.2
344-1413H1	14.4	10.1	23.1	4.9	38.8	8.7
496-H2	10.2	5.4	21.8	28.9	10.5	23

Table 1 The percentage of Heating adoption clusters for each circuit in Station 496 Hyde Park

Circuit	EV Cluster I (in %)	EV Cluster II (in %)	EV Cluster III (in %)	EV Cluster IV (in %)	EV Cluster V (in %)
344-1411H2	0	5.7	17.1	30	47.1
441-1401H1	0	7	47.3	28.7	17.1
496-H3	19.3	11.5	24.4	28.7	17.1
496-H5	31.3	15.2	28.5	12.2	12.9
496-H7	15.8	31.3	17	20.5	15.4
60-1409J1	14.1	19.6	14.6	27.1	24.6
344-1411H1	0	25	50	12.5	12.5
496-H4	16.7	9.9	21.3	22.5	29.5
496-H8	7.6	25.5	20.3	29.6	16.9
344-1412H1	8.3	23.3	28.1	35	5.3
362-1207H1	29	38.6	13.9	16.5	2
362-1209H1	15.2	30.4	22.5	27.7	4.1
496-H6	34.5	15.5	27.3	8.9	13.7
496-H1XY	7.5	28.3	19.5	22.5	22.3
344-1413H1	6.3	18.1	27.7	34.6	13.3
496-H2	10.7	16.3	17.6	20.4	35

Table 2 The percentage of EV adoption clusters for each circuit in Station 496 Hyde Park

The above pie chart and tables show the EV and Heading adoption clusters by year 2050 per state predictions.

Overview of the Area Electric System

Transmission System Overview

The # 496 Hyde Park Substation is stepped down from 345kV to 115kV via one (1) transformer and from 115kV to 13.8kV via two (2) transformers to supply the Substation load. Two (2) 115kV lines, 496 – 528 and 496 – 529 tie into the # 496 Hyde Park Substation from #110 Baker Street Substation. The 345kV line 3164 to #496 Hyde Park is fed from #330 Stoughton Substation

Hyde Park 345kV steps down to 115kV and ties into two separate Hyde Park 115kV Buses. The 115/13.8kV transformer 110B and the 115kV circuit 496-529 from #110 Baker Street Station ties into one of the buses. The 115/13.8kV transformer 110A and the other 115kV circuit 496-528 from #110 Baker Street Station ties into the other bus.

The 115kV circuits from #110 Baker Street Station, 496-528 and 496-529, are 3500 AL pipe type cables. The 345kV circuit 3164 is a 2500CU HPFF cable.

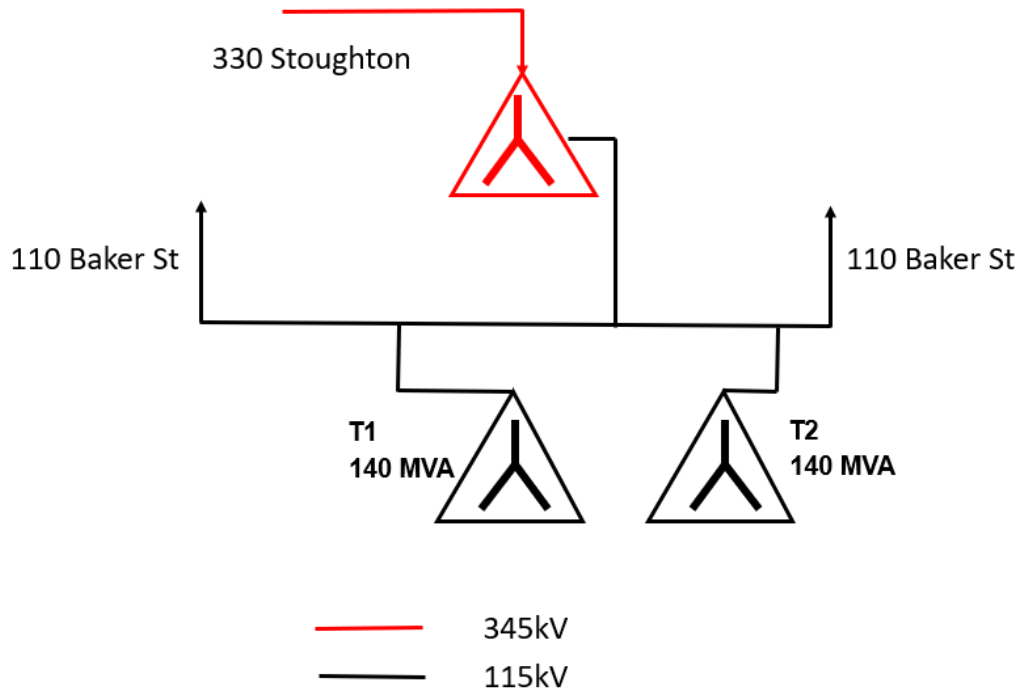


Figure 4 Hyde Park Substation (345kV and 115kV)

An N-1 on any of the Hyde Park 115kV buses or 115/13.8kV transformers leads to the Hyde Park load being supplied by one Hyde Park 115/13.8kV transformer. This causes a thermal constraint at the Hyde Park Substation as the Hyde Park load would not be able to be carried with a single transformer. This is covered under the Capacity Deficiency section.

Distribution System Overview

There are four (4) Hyde Park 13.8kV buses tied via a normally open breaker. One 115/13.8kV transformer supplies two of the buses and the other transformer supplies the remaining two. The two transformers were installed in 1973 and neither of these transformers currently exhibit asset health issues. Each transformer has a rating of 146 MVA, which is also the firm capacity of the station. The Substation has a load-carrying capability (LCC) of 159 MVA, which accounts for the loss of one of the 115/13.8kV transformers plus transfer capability to other bulk distribution Substations.

An N-1 on any of the Hyde Park 115kV buses or 115/13.8kV transformers leads to the Hyde Park load being supplied by one Hyde Park 115/13.8kV transformer. This causes a thermal constraint at the Hyde Park Substation as the Hyde Park load would not be able to be carried with a single transformer. This is covered under the Capacity Deficiency section.

There is currently approximately 16.9 MW of solar and battery at Hyde Park Station. The four 13.8kV Hyde Park Substation feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H that the BESS would be connected to are supplied from four different Hyde Park 13.8kV buses.

1. Circuit 496-H1XY has approximately 0.4 MW of solar and is supplied from Bus 1.
2. Circuit 441-1401H has 1.7 MW of solar and is supplied off of Bus 2.
3. Circuit 362-1207H has approximately 0.7 MW of solar and is supplied off of Bus 3.
4. Circuit 362 – 1209H has approximately 0.9 MW of solar connected and is supplied off of Bus 4.

There is approximately 6095.3 kW of solar in queue at Hyde Park substation, including approximately 58.7 kW of solar in queue on circuit 496-H1XY and 79.5 kW of solar in queue on circuit 362-1209.

The Hyde Park 13.8kV feeder 496-H1XY supplies 1,700 customers, 362-1207H feeder supplies 107 customers, 362-1209H feeder supplies 320 customers and 441-1401H feeder supplies 1,114 customers.

Below is an image showing the four feeders, which are color coded and labelled on top of the image.

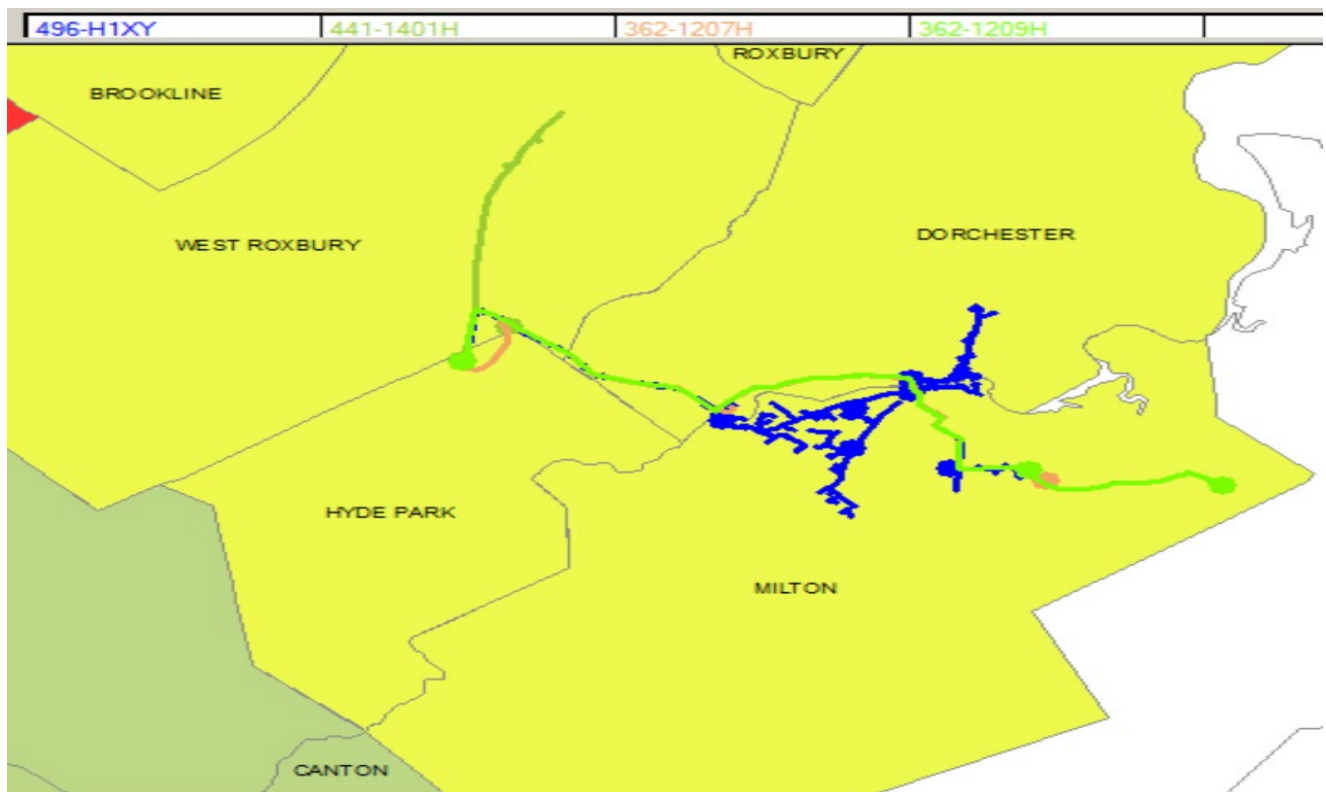


Figure 5 Hyde Park 13.8kV Feeder Diagram

There are twenty-four 13.8kV feeders supplying Hyde Park Substation load, six feeders tied to each of the four 13.8kV buses.

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The following table shows the firm capacity and Load Carrying Capability of the Hyde Park Station for the next ten years.

2023					2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Year	Station Number	Station Name	Total Station Transformer Capacity (MVA)	Firm Capacity	Load Carrying Capability									
2023	496	Hyde Park	280	146	159	158	158	157	157	156	156	156	155	155

Table 3 LCC of Hyde Park (MVA)

Distributed Energy Resources Overview

Circuit 496-H1XY has approximately 0.4 MW of solar and is supplied from Bus 1. Circuit 441-1401H has 1.7 MW of solar and is supplied off of Bus 2. Circuit 362-1207H has approximately 0.7 MW of solar and is supplied off of Bus 3. Circuit 362 – 1209H has approximately 0.9 MW of solar connected and is supplied off of Bus 4. The total amount of solar connected to feeders supplied by Bus 1 is 10 MW and by Bus 2 is 15.6 MW. The total amount of solar connected to feeders supplied by Bus 3 is 3.4 MW and by Bus 4 is 3.7 MW. There is approximately 6.1MW on solar in queue at Hyde Park substation. There is approximately 58.7 kW of solar in queue on circuit 496-H1XY and 79.5 kW of solar in queue on circuit 362-1209. This compares to the load forecasted impact of solar in Table 7 (further below) of 0.34 MW of solar in year 2032 for Station 496 Hyde Park. The Combination of the evening peak between 6 pm and 8 pm as well as the firm weather adjustment contribute to the discrepancy between the actual installed and forecasted peak impact. The evening peak is between 6pm to 8pm, the available sunlight is low during these hours. Therefore, the load reduction is small. An adjustment factor for the irradiance and weather is applied. Additionally, the load forecast reduction in Table 7 is not the reduction for "total connected PV". It is the reduction for 'additional PV' from today and excludes existing PV, (the reduction from which is already captured in the base load).

The solar contribution depends on the time of day and weather and the load reduction using solar would be miniscule. Therefore, Eversource cannot count on solar to address our peak load overload concern.

Load Forecast

Hyde Park Substation supplies 61,740 customers in Hyde Park and Milton. The Hyde Park 13.8kV feeder 496-H1XY supplies 1,700 customers, 362-1207H feeder supplies 107 customers, 362-1209H feeder supplies 320 customers and 441-1401H feeder supplies 1,114 customers. These circuits supply approximately 5.2% of customers supplied by the station.

Hyde Park Substation 2021 peak load was 157MVA, which exceeded the 2021 Load Forecast by 7MVA; It is 108% or 11MVA above Substation firm capacity of 146MVA; 98.7% of the Substation Load Carrying Capability (LCC) of 159 MVA. In 2021, 7.7 MVA was transferred

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permanently off of Hyde Park Substation after the Substation peaked for that year. The 2022 post load transfer peak load was still 152MVA, 104% of the Substation firm capacity of 146MVA and 95.6% of the Substation LCC of 159 MVA. About an additional 5 MVA of load was transferred in 2022 after the Substation peaked for that year. In 2023, the substation was loaded to 97% of its firm capacity. A total of 13 MW of load was transferred to nearby Substations using existing field switches. However, even with these load transfers, the Hyde Park Substation is forecasted to be about 95% of the Hyde Park substation LCC by 2024. The Hyde Park transformers are forecasted to be above the transformer LTE rating of 146 MVA under an N-1 contingency by the summer of 2024. This violates the Distribution Planning criteria. Therefore, there is a critical need for a solution to address thermal capacity constraints at Hyde Park Substation.

Building a new Substation to address this thermal constraint will take a number of years and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal Substation constraint until the long-term plan is in service.

Below in Table 4 is the Hyde Park forecast with a total of 13 MW of load transferred from Hyde Park Substation to nearby Substations. The 13 MW load transfer was implemented by using existing field switches. However, such transfer reduces the existing circuit transfer capability, increases the exposure to the existing distribution lines and reduces the spare capacity at the receiving Substation.

2023					2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Year	Station Number	Station Name	Total Station Transformer Capacity (MVA)	Firm Capacity	Load Carrying Capability									
2023	496	Hyde Park	280	146	159	158	158	157	157	156	156	156	155	155

(From Table 3) LCC of Hyde Park (MVA)

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
496	Hyde Park	Forecast Load	150.83	150.86	150.97	151.09	151.36	151.81	152.43	153.41	155
		Load/LCC	94.9%	95.5%	95.6%	96.2%	96.4%	97.3%	97.7%	98.3%	100%

Table 4 Hyde Park Load Forecast (MVA)

The load transfers incorporated in the LCC are again shown below in Table 3. Under emergency conditions, the following switching procedures occur to transfer load via distribution ties: 17.6 MVA or 735 Amps of load can be transferred to nearby substations. The four highest load transfers that utilize remote controlled switches (the switching procedures in Table 5 - rows 2,3,4 and 6) would occur. The switching procedures marked with an asterisk are the ones that use remote controlled switches. This amounts to about 550 A or 13.2 MVA of load transfer, which is the amount that is added to the firm capacity of the station to obtain the LCC. If

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required, the remaining load (two switching procedures) would also be transferred via manual switches. The circuits in the below table are tied via reclosers, automatic sectionalizing units or manual load break switches.

The switching would occur in under a minute and the customers typically do not see outages during this switching. The distribution system is designed to limit outages when going to off schedule conditions.

FROM CIRCUIT	TO CIRCUIT	LOAD	CLOSE	OPEN
496-H1XY	483-H4 (STA 483)	70 Amps	* ASU Switch on P.104/143 Adams St, DOR "check phase shifters"	* ASU switch on P.43/14 Eliot St, MIL
496-H8	483-H7 (STA 483)	100 Amps	* ASU Switch on P.1882/10 Ashton St, DOR "check phase shifters"	* ASU switch on P.1119/1 Austin St, DOR
496-H4	284-1430H (STA 350)	125 Amps	* Recloser on P.59/19 Brookside Ave, WRX "check phase shifters"	* PME-9, PMH24645 side B Washington St, WRX
496-H2	470-H9 (STA 470)	100 Amps	* Recloser on P.26/6 Milton St, MIL	* Recloser on P.15/11 Neponset Valley Pkwy, MIL
496-H3	20-H7 (STA 20)	115 Amps	Manual load break on P.58/4 Laurie Ave, WRX	* Recloser on P.112//7 Gardner St, WRX
344-1412H1	20-H3 (STA 20)	225 Amps	* ASU switch on P.130/1 Sprague St, DED	* Recloser on P.130/6 Sprague St, DED

Note: ASU = Automatic Sectionalizing Unit, Dorchester = DOR, Milton = MIL, Dedham = Dedham, West Roxbury = WRX

Table 5 Hyde Park Emergency Load Transfers

The load is transferred to Baker St, Colburn St, Dewar St and Canton St of which the first three stations are also heavily loaded and are more than 90% of their LCC by 2025, making the need for the BESS even more critical. The forecast for these stations is shown in the below table. Furthermore, two of the circuits which are proposed for the BESS interconnection, 496-H1XY and 362-1209H run up to 90% of their summer normal ampacity. The BESS would help to alleviate the load on these heavily loaded circuits as well.

	Station	Total Station Transformer Capacity (MVA)		2024	2025	2026	2027	2028	2029	2030	2031	2032
110	Baker Street	150	Forecast Load Load/LCC	80 95%	80 95%	80 95%	80 95%	81 95%	81 96%	81 96%	81 96%	82 98%
350	Colburn Street	261	Forecast Load Load/LCC	179 84%	192 91%	196 93%	196 93%	196 93%	197 94%	197 94%	198 95%	199 95%
470	Canton	196	Forecast Load Load/LCC	82 54%	82 55%	81 54%	81 54%	81 54%	81 54%	82 55%	83 55%	84 56%
483	Dewar Street	280	Forecast Load Load/LCC	143 99%	142 99%	142 99%	142 99%	142 99%	142 99%	143 100%	144 101%	146 102%

Table 6 Load Forecast for Tie Stations

Table 7 below is the breakdown of the Hyde Park summer forecast components. Hyde Park Station 496 services a relatively large area with a significant residential component to that load. This, and limited land availability, as well as the evening peak and weather adjustment of solar output yield very low solar compensation at the station. The 'PV' column shows the contribution of solar; the 'EV' column shows the Electric Vehicle charging contribution and 'EE' column shows the Energy Efficiency contribution.

Forecast Year	Trend (MW)	PV (MW)	EV (MW)	EE (MW)	Load Transfers During Emergency (MW)	Step Load (MW)	Gen. (MW)	Final (MW)
2023	145.15	-0.01	0.03	-0.47	13.17	1.0	0.0	145.7
2024	145.95	-0.03	0.10	-1.29	12.6	6.1	0.0	150.83
2025	146.66	-0.06	0.23	-2.07	12.1	6.1	0.0	150.86
2026	147.33	-0.09	0.43	-2.80	11.7	6.1	0.0	150.97
2027	147.92	-0.13	0.74	-3.53	11.2	6.1	0.0	151.09
2028	148.52	-0.17	1.17	-4.27	10.7	6.1	0.0	151.36
2029	149.19	-0.21	1.74	-5.00	10.3	6.1	0.0	151.81
2030	149.83	-0.26	2.48	-5.73	9.9	6.1	0.0	152.43
2031	150.49	-0.30	3.60	-6.47	9.5	6.1	0.0	153.41
2032	151.22	-0.34	5.23	-7.20	9.1	6.1	0.0	155.0

Table 7: Company Forecast for Station 496 Hyde Park issue Q1-2023 with no adjustments

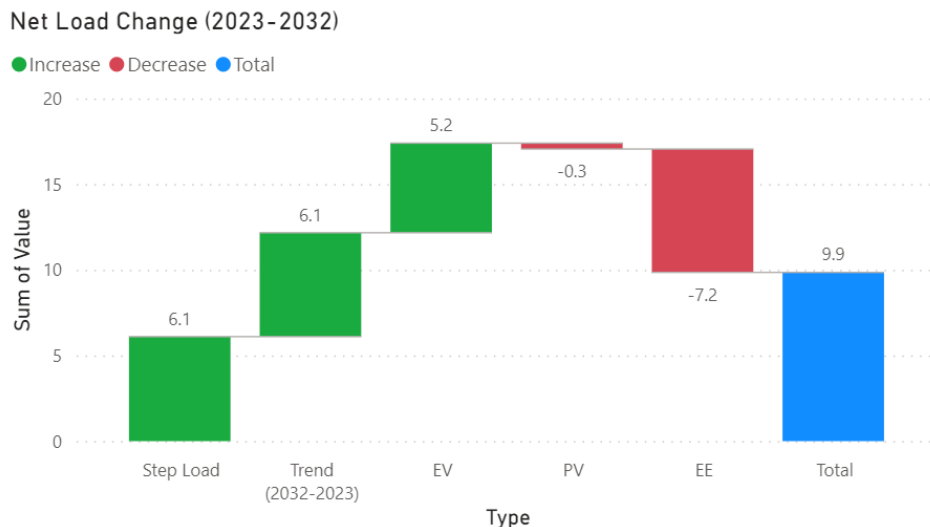


Figure 6: Net Load Change at Station 496 Hyde Park based on Company Forecast issue Q1-2023

The step loads are residential as well as for EV charging station and services.

During the next 10 years, there is step load and a steady increase in EV load that contributes to the continuous load growth at Hyde Park substation.

In the long term, much like most of the Company's Territory, a larger scale transition to electric heating is likely to occur, especially in a highly residential area with dense urban/commercial settings. With a transition to Winter Peak, heating will dominate the peak load at Station 496, which, under full electrification potential (EV and Heating), is projected to hit 573 MW. Figure 7 below highlights the makeup of the then morning peak.

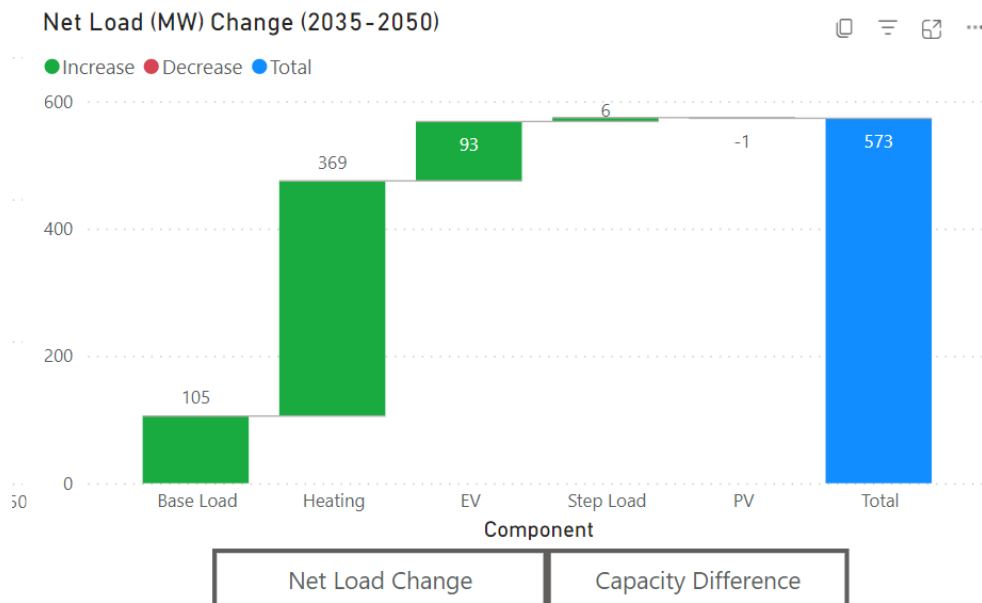


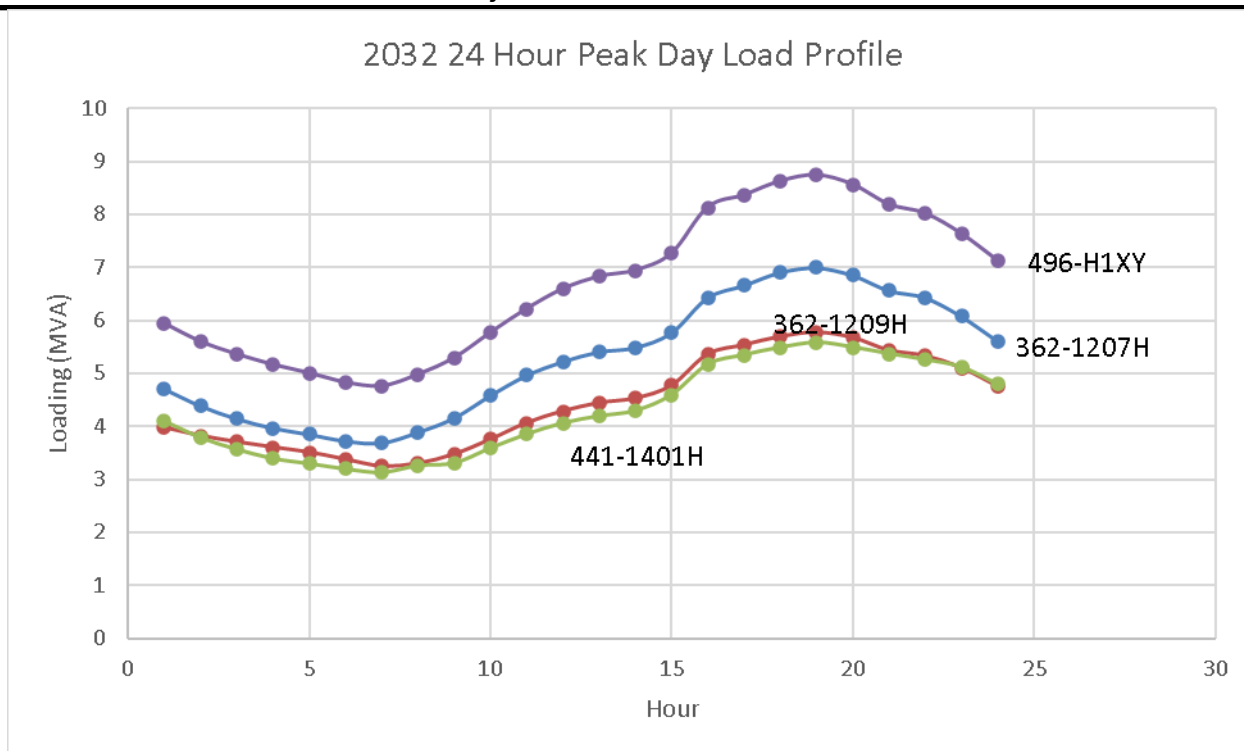
Figure 7: Net Load Composition Station 496 Hyde Park ESMP Long Term Assessment

Distribution Feeder Capacity

Two of the four feeders have capacity issues: the Hyde Park circuit 496-H1XY supplied from Bus 1 historically runs 80-90% of its summer normal ampacity in the summer. The circuit 362-1209 supplied from Bus 4 historically runs 60-90% of its summer normal in the summer months. The BESS would help relieve load on these heavily loaded distribution circuits as an added benefit. The function of the BESS will be described more in detail in the below sections discussing the alternatives.

Criteria: Feeder upgrades are required when [80% of normal feeder rating for cables or 90% normal rating or emergency for wires] is violated to ensure that any feeder cable/wire will not exceed normal or emergency ratings, as per Section 3.1 [of the DSPG]. (Section 2.10, DSPG 2020)

The below figure shows the forecasted 2032 distribution feeder load for the four Hyde Park 13.8kV distribution feeders that would connect to the BESS, 496-H1XY, 441-1401, 362-1209 and 362-1207.



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Figure 8 2032 Hyde Park 13.8kV Feeder Loading

Table 8 shows the forecasted 2032 loading on the four Hyde Park 13.8kV feeders compared to their ratings.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 PM	6.7	5.5	5.3	8.4
6:00 PM	6.9	5.7	5.5	8.6
7:00 PM	7.0	5.8	5.6	8.7
8:00 PM	6.9	5.7	5.5	8.6
9:00 PM	6.6	5.4	5.4	8.2
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Table 8 2032 Hyde Park 13.8kV Feeder Loading

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[2] PROJECT OBJECTIVES

System Planning Project Objectives

Hyde Park 115/13.8kV substation peak load is forecasted to be above the firm capacity of the station and above 90% of the LCC of the station. The load continues to increase due to step load and EV electrification load. Project 21192 Hyde park-Dorchester Area Supply Initiatives will allow the construction of a new substation, which will address this capacity constraint. However, building a new Substation to address the thermal constraint will take a number of years and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal Substation constraint until the long-term plan is in service. The objective is to mitigate capacity constraints at Hyde Park for the short term to allow time for a longer-term solution to be developed.

[3] Alternatives Considered with Cost Estimates

Introduction of Solution Alternatives

Below is the Hyde Park forecast with a total of 13 MW of load transferred from Hyde Park Substation to nearby Substations.

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
		Forecast Load	150.83	150.86	150.97	151.09	151.36	151.81	152.43	153.41	155
496	Hyde Park	Load/LCC	94.9%	95.5%	95.6%	96.2%	96.4%	97.3%	97.7%	98.3%	100%

(From Table 5) Hyde Park Load Forecast

The load is predicted to be about 95 percent of LCC in the summer of 2024 and will continue to increase. The nature of the load increase at stations can be unpredictable. The need for a short-term solution is urgent until a long term solution is in place.

Non-Wires Solution Alternatives

A pre-screening assessment was performed based on the scope identified. Projects that answer 'No' to one or more item of the questionnaire do not need to have a non-wires alternative (NWA) solution screening performed. As shown in the below table, all responses are 'Yes' and an NWA screening has been performed.

NWA Pre-Screening Criteria	Response
Preferred Solution Capital Cost > \$3M	Yes
Capacity Need > 3+ Years	Yes
Main Project Driver is Capacity	Yes

Table 9. Non-Wires Alternative pre-screening assessment

Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any load relief solution.

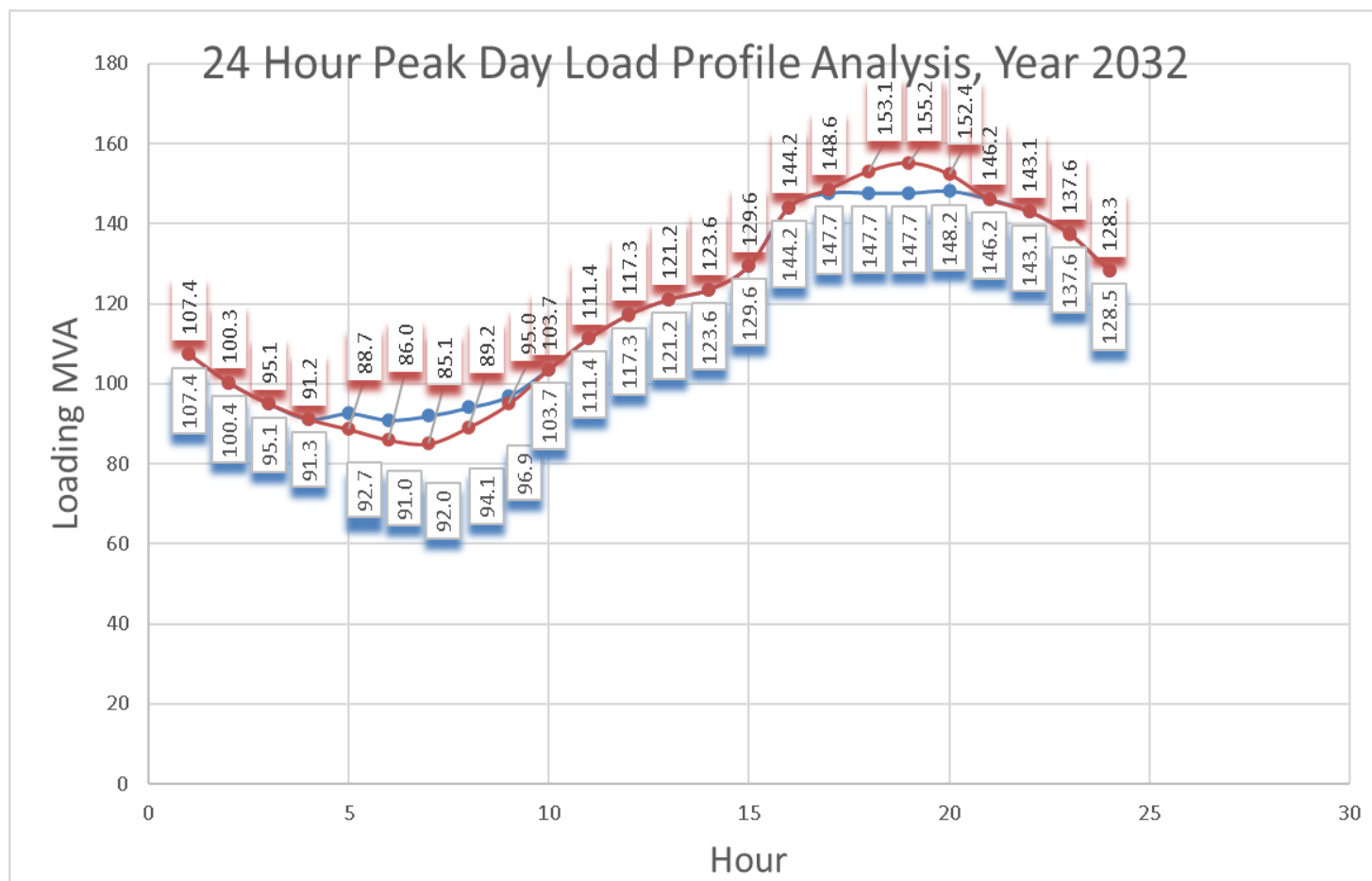


Figure 9 2032 Hyde Park Substation Loading

In figure 9, the red line represents the Hyde Park Substation load without any load relief solution and the blue line represents the Substation load with 10MW/20MWH batteries installed. The batteries are seen to be acting as a load when they charge from 4:00 AM to 9:00 AM and they

relieve substation load when they discharge from 17:00PM to 21:00PM. Four 2.5MW/5MWH batteries are chosen to allow for bus and feeder diversity as well as battery diversity. As shown in figure 10 below, each battery is tied to a feeder supplied from a different Hyde Park 13.8 kV Bus. If there is a fault on either the bus or feeder, the BESS can connect to the remaining buses and feeders. Similarly, if there is a failure in one of the batteries, there would still be 7.5MW/15MWH capacity available from the batteries.

According to the NWA analysis, the benefit cost ratio of a BESS to traditional alternatives is 0.67 implying that the traditional alternatives would be more favorable from a cost perspective. However, it is critical to note the hurdles and timing delay that would be faced if the traditional alternative were to be implemented as explained in the alternatives in sections below. The traditional alternative would be to construct two 13.8kV feeders from Needham station to nearby Hyde Park 13.8kV feeders. Opposition in Needham is expected since a project was just completed through Needham. There would also be potential hurdles in gaining rights and permits. This alternative may take significantly longer than the BESS to implement. This is explained in detail in the alternatives section.

Energy Efficiency (EE) Screening

EE offers continuous savings across the entire load profile rather than focusing on peak periods. It is important to note that the EE impacts, which appear to be considerable in terms of annual energy savings, correlate to relatively low peak contributions. Because advantages are spread out throughout the entire year and the impact is not favorable for delaying a system upgrade. Consequently, while Energy Efficiency is not a workable solution on its own, it can be combined with other solutions.

Demand Response Screening

As demand response does not constitute a reduction of demand, but rather a shift of demand to a different time, highly saturated stations such as Hyde Park do not make good candidates for DR programs as these programs will cause, with their snap back effects (high load peaks after a DR response) the peak to shift later to the evening with even less as already accounted for solar offset. Further, DR contracts have opt out clauses making them less firm of a resource, and limitations on the number of events that can be called.

Storage Screening

For Hyde Park Substation a storage capacity analysis was conducted, yielding the required power rating and battery capacity to fully address the forecasted need. Per the above graph, the batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load. In year 2032, the Hyde Park substation peak load would be 101% of firm capacity and 95.3% of LCC with a 10 MW/20MWH BESS installed. **This makes this solution the most technically feasible NWA solution.**

Solar PV Screening

Due to the uncertainty around solar output based on weather conditions, the weather adjusted capacity profiles considered for Hyde Park and based on the NWA Framework are not sufficient to reliably offset the capacity need.

Combined Heat and Power Screening

Given current political settings in the Commonwealth and the City of Boston specifically, the Company does not pursue the permanent installation of combustion solutions as a viable offset to traditional infrastructure.

NWA Solution Proposal

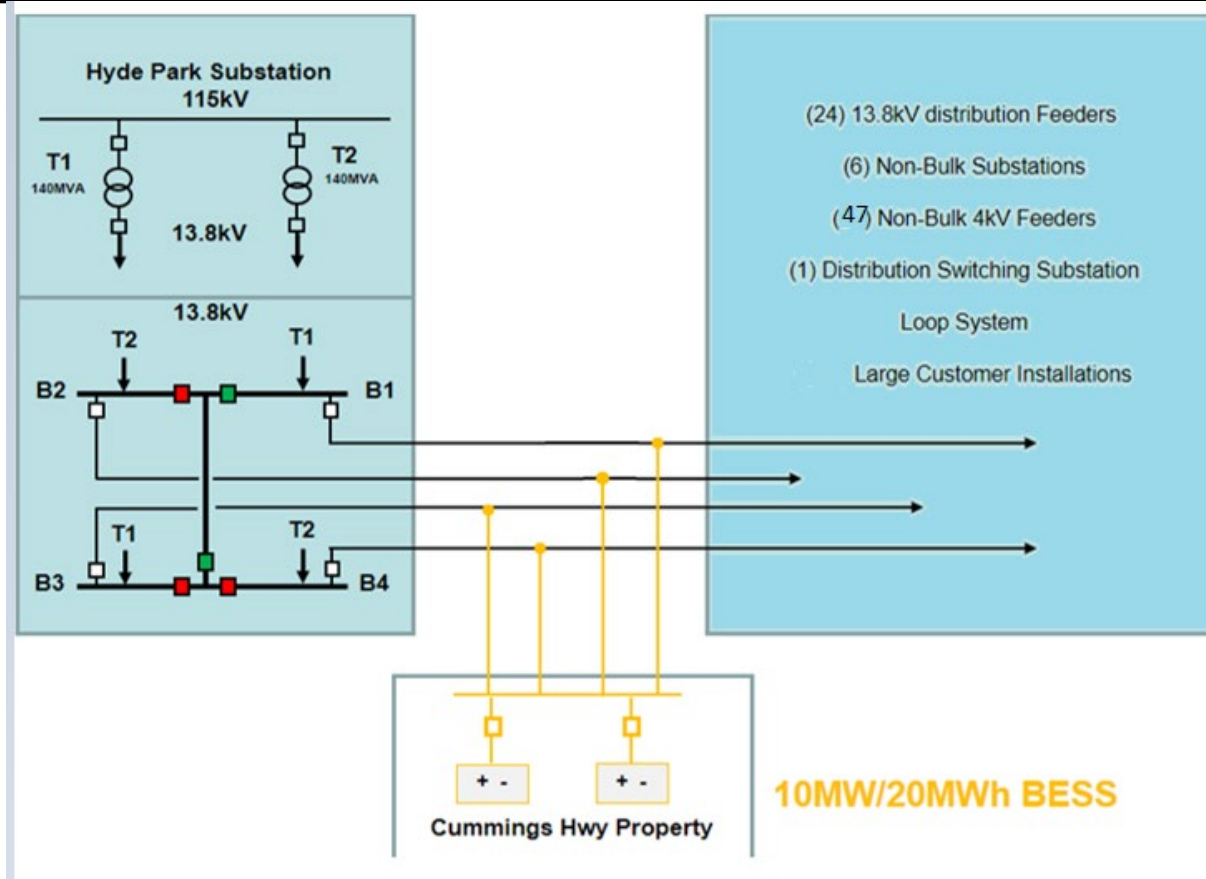
Due to the frequency of the event days and the magnitude of the capacity need as well as the results of the above NWA screening, the only technically feasible option would be the BESS.

Alternative 1 – Hyde Park BESS Solution

After considering multiple solution alternatives including Non-Wires Alternatives (NWA) solutions and other Distribution only solutions and using historical data and engineering models including PSCAD and Synergi analysis, the optimal solution to address the totality of the area needs would be to install four (4) 2.5MW/5 MWh battery units at 427 Cummins Highway in Hyde Park Massachusetts, each connected to a Hyde Park 13.8Kv distribution feeder. The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the Substation peak. The substation peak exceeds the firm capacity of the station and is forecasted to keep rising due to step loads and EV charging load. The Battery can ride through multi-day peak load events. The four (4) 13.8Kv distribution feeders would be 496-H1XY, 441-1401, 362-1209H and 362-1207H. Some of those feeders benefit from the BESS as they historically run up to 90% of their summer normal ampacity. The scope will be fully described in Section 4 Project Scope. BESS can be connected to Baker St#110 side in the future through 25Kv or 4Kv feeders to support Baker Substation if needed, which will require underground cable construction.

The below diagram shows the Hyde Park station configuration as well as the proposed BESS connection.

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Figure 10 Hyde Park Distribution System and Proposed BESS

When analyzing the above solution, the system planner(s) performed both dynamic and steady state analysis to determine the sizing and system needs. To complete the steady state analysis, the substation model had to be validated and studied to determine the charging schedule as well as the most efficient way the BESS can be used to address system needs. The dynamic study performed for this project is unique to distribution substation studies because dynamic studies are typically performed on individual feeders or on the transmission system with a simplified distribution topology/conversion. Since this is a unique study, a model using PSCAD software was built with the substation, feeders, and BESS solution had to be built and contingency analysis performed. PSCAD is typically used to perform dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis. Due to the complexity of the model, the effort took three months to complete.

Figure 9 shows that the four (4) 2.5MW/5 MWh battery units can reduce the peak load to approximately 95% of the Load Carrying Capacity of the Hyde Park Substation in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 13.8Kv distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders.

The project is designed to meet the 10 year forecast, where the highest load being is in 2032. Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any batteries installed as seen in figure 9. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load.

The red line in figure 9 represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

Below graph shows the 2032 24-hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational. It shows the 24-hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load.

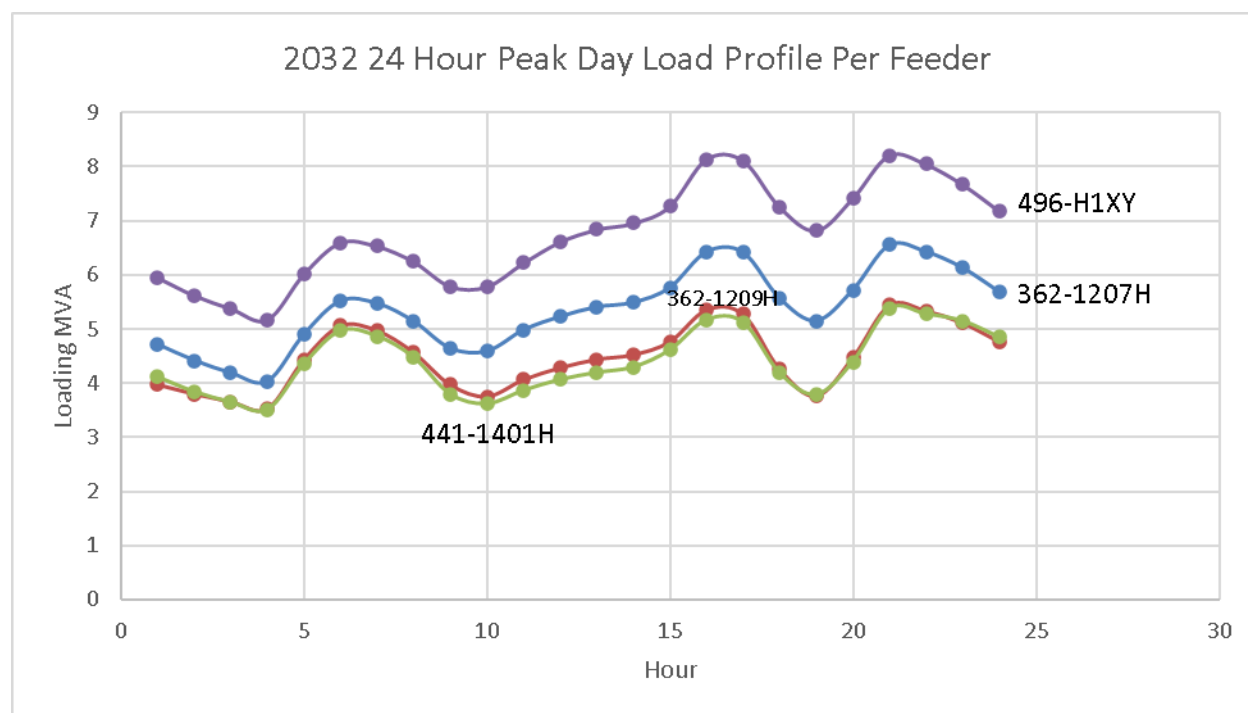


Figure 11 2032 Hyde Park 13.8kV Loading

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 04:00 AM to 09:00 AM and acting as a load. System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.9	4.4	4.4	6.0
6:00 AM	5.5	5.1	5.0	6.6
7:00 AM	5.5	5.0	4.9	6.5
8:00 AM	5.1	4.6	4.5	6.3
9:00 AM	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Table 10 2032 Hyde Park 13.8kV Feeder Loading With BESS

The exact loading for the year 2032 cannot be predicted at this stage. However, based on the above loading data, there is a flexibility to charge the BESS on three of the above feeders instead of four if need be. Based on the above loading, feeder 441-1401 has the lowest amount of load and would be able to accommodate another 2.5 MW of BESS while feeder 496-H1XY may not. During the annual planning forecast, System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decreased load of 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 PM is due to the BESS. The batteries are seen to be acting as a load when they charge from 4:00 AM to 9:00 AM and they relieve substation load when they discharge from 17:00PM to 21:00PM. The red line represents the Hyde Park Substation load without any load relief solution and the blue line represents the Substation load with 10MW/20MWH batteries installed.

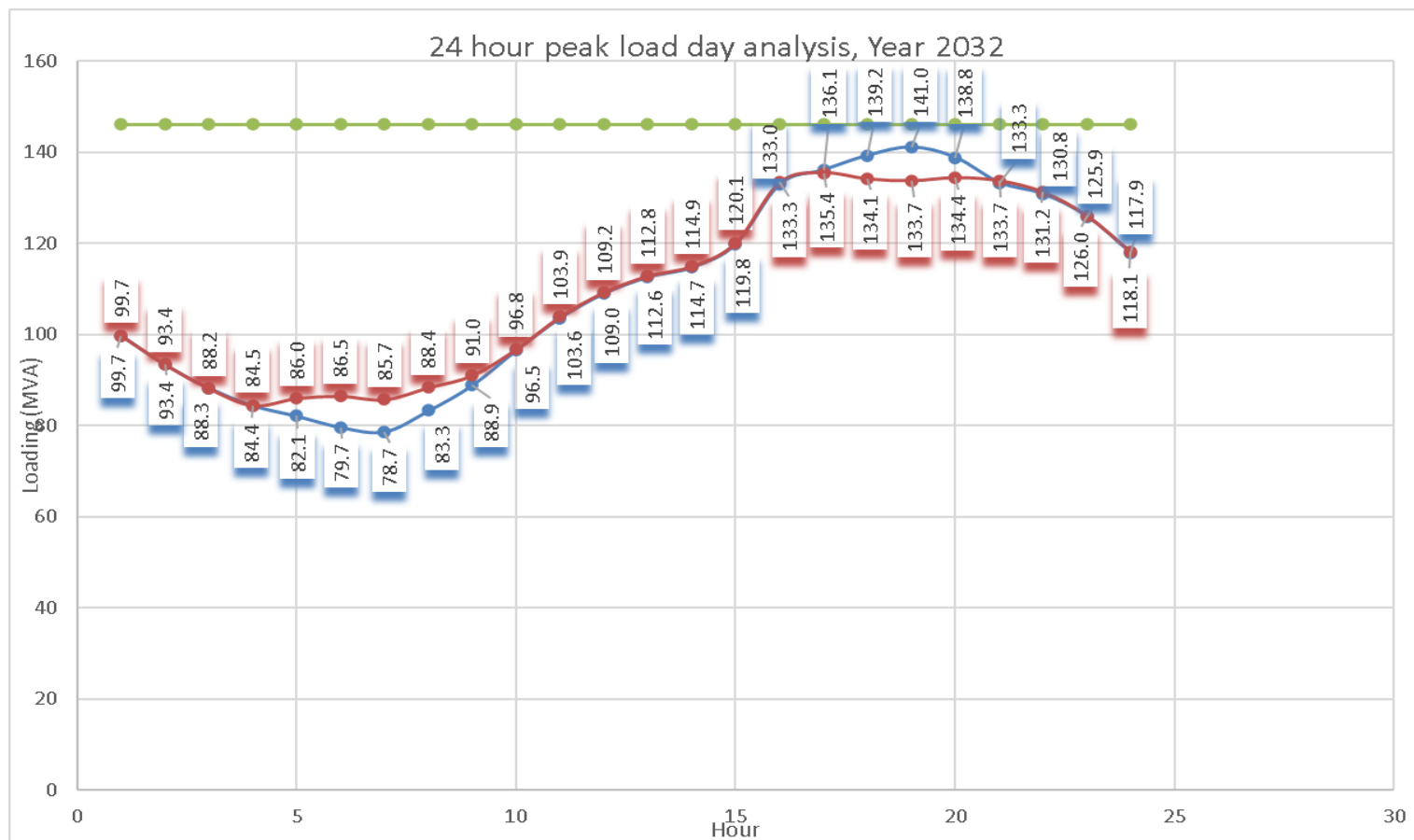


Figure 12 2032 Hyde Park Substation 24 Hour Loading Under N-1

Below is a graph showing circuit loading for each of the four feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four feeders. It shows the 24-hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load.

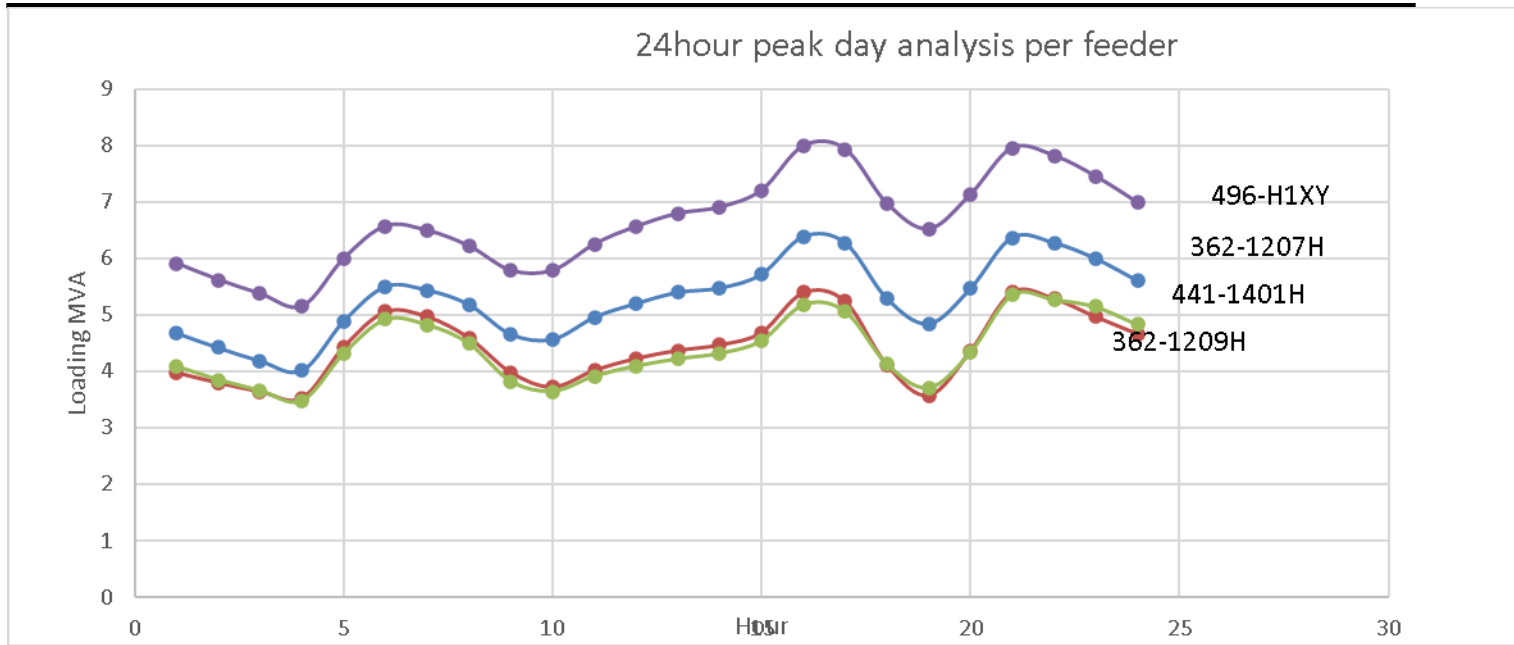


Figure 13 2032 Hyde Park 13.8kV 24 Hour Feeder Loading Under N-1

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 04:00 AM to 09:00 AM and acting as a load.

	Feeder 362 1207H	Feeder 362 1209H	Feeder 441 1401H	Feeder 496 H1XY
5:00 AM	4.89	4.43	4.32	5.996
6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Table 11 2032 Hyde Park 13.8kV Feeder Loading With BESS Under N-1

The BESS also helps to bring down the peak load on these 13.8 kV feeders. The Hyde Park circuit 496-H1XY historically runs 80-90% of its summer normal ampacity in the summer. Circuit 362-1209 historically runs 60-90% of its summer normal in the summer months.

To study the impact of the BESS on the Hyde Park distribution system, a transient dynamic analysis of the preferred alternative was performed in the PSCAD (Power Systems Computer Aided Design) tool. The four Hyde Park 13.8kV distribution feeders along with the proposed

BESS systems were modeled¹ in PSCAD by using circuit load and impedance data collected from steady-state Synergi models. All other distributed generation (DG) on the four distribution feeders were modelled with a generic inverter model, not specific to any one manufacturer. The rest of the DG connected to the Hyde Park station was modeled as a lumped generation at the 13.8kV bus. The dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis were done to verify compliance with IEEE 1547-2018 Standard.

The ROI analysis determined the BESS inverter's response to unintended island conditions. According to the IEEE standard, for an unintended islanding condition, the distributed energy resource (which in this case is the BESS) shall detect the island, cease to energize the island, and trip within 2 seconds of the formation of the island.

When interconnecting distributed energy resource there is a concern that during step changes in load (such as tripping of an upstream device), the proposed inverters may cause transient over voltages in excess of 1.2 per unit, which can potentially cause damage to the customer's equipment and/or utility equipment. The TOV analysis was conducted to verify if the expected transient overvoltage conditions caused by the BESS inverters during step changes in load were within standard limits.

The ROI and the TOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios. The N-1 operational scenarios included one of the four feeder outages, a transformer outage, or a bus outage. The BESS system passed the ROI test for all scenarios tested. The BESS system also does not contribute to any TOV conditions for any of the tested scenarios.

Alternative 2 - New 13.8kV Feeders From Needham Substation to Hyde Park 13.8kV Feeders

Bring additional capacity to Hyde Park area by constructing two 13.8kV distribution lines from Needham Substation.

The Needham station firm capacity is 142 MVA while the projected peak load in the next ten years is 88 MVA.

Based on an assumption that the new dedicated lines from Needham Substation#148 will run 100% underground path not currently occupied by common duct. The two new lines would run from Needham Station to nearby Hyde Park 13.8kV circuits, 496-H3 and 496-H6. The need to provide these two circuits would require:

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit

21 (55 foot poles)

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8kV circuit 496-H3:

19,900 feet of 3-700FS

¹ The BESS systems on the distribution circuits were modelled with a generic inverter model, not specific to any one manufacturer. After a vendor for the BESS is selected, the dynamic study will have to be re-run with the manufacturer specific inverter model.

2,450 feet of 477 ACSR
2 New Reclosers
2 New Underslung
2 new 45 foot poles

This new circuit would offload 7MVA from this 496-H3 circuit.

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8kV circuit 496-H6:

25,900 feet of 3-700FS
2,300 feet of 477 ACSR
2 New Reclosers
2 new Underslung

This new circuit would offload 6.21MVA from this 496-H6 circuit.

There is some load that would still be kept on 496-H6 and H3 just for cable monitoring purposes so that the cable will have readings and can therefore be utilized under a fault via a tie switch.

With this alternative, the total possible capacity on each circuit is 11.5MVA per circuit.

The below diagram shows the high-level route from Needham Substation to Hyde Park 13.8kV circuits 496-H3 and 496-H6.

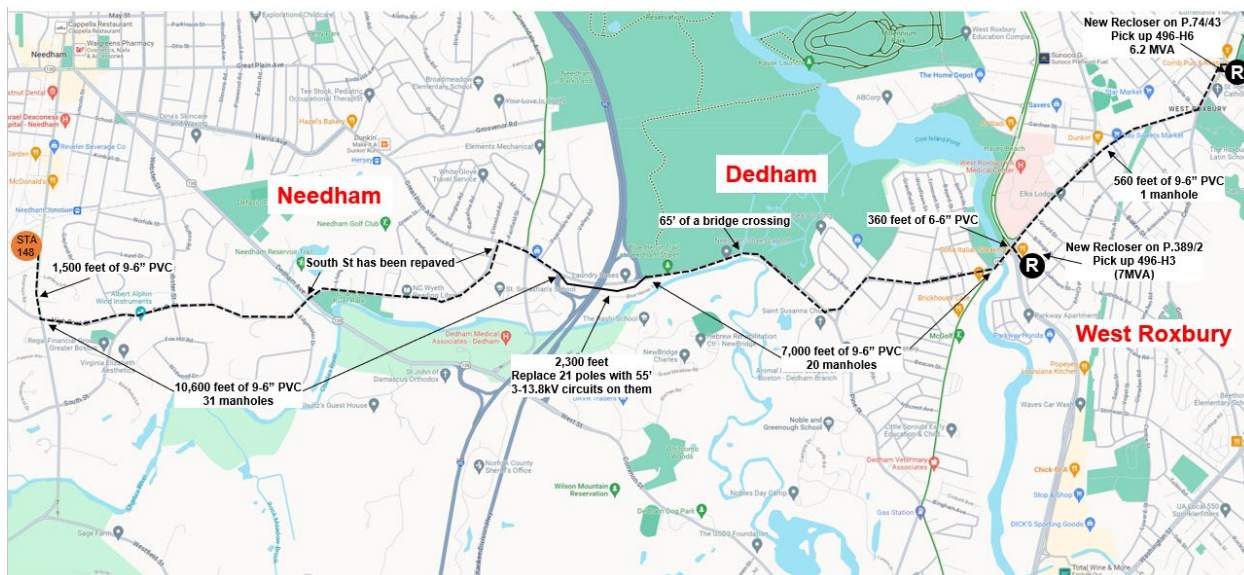


Figure 14 New Needham 13.8kV Circuits

The Hyde Park 13.8kV feeders are not relieved completely for cable monitoring purposes. If there is no load left on the feeders, they cannot be monitored to determine whether they can be utilized as distribution ties.

By July 2027, the capacity to be delivered by the project 21334 will need to be delivered by other means. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours typically from 6 PM to 8 PM. Based on the 2023 load

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forecast if the BESS is not installed by the need date of 2027, two generators of one MW capacity would be needed in 2027 and 2028 and two generators of two MW capacity would be needed in 2029 until the BESS is installed. 8 MW of generation would be needed in the worst-case scenario where the BESS does not get installed until the station is at 100% of LCC. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time. System Planning will monitor the annual forecast that may shift the plan laid out above.

Alternatively, the option of running the 13.8kV circuits through an overhead right of way to connect to the two Hyde Park 13.8kV circuits was also analyzed. This option was not feasible because the railroad easement along the path would limit the number of circuits that can run along the route and the width between the tracks and edge of easement is also too limited to accommodate two 13.8kV circuits in some locations.

Alternative 3 - River Street BESS Solution (Alternative Considered but deemed not viable)

The City of Boston proposed installing 10MW/20MWH BESS at 201 River St property in the City of Boston. The BESS is proposed to be interconnected to adjacent 4kV substation to help relieve upstream loading of the feeders and to be utilized for micro-grid purposes for the nearby City campus.

The property does not meet the projects functional and electrical requirements as it relates to the capacity needs of Hyde Park substation. The preferred alternative is meant to interconnect to the four distribution circuits mentioned in the Hyde Park BESS solution, which essentially feeds blocks of customers. So, moving the battery storage to the river street location minimizes our ability to feed and assist those four blocks of customers off of those four separate circuits. The river street location is downstream of a 4kV Step down substation or the #311 substation. The River St location would certainly require extended or additional distribution upgrades that were not accounted for because the interconnection would be on the 4kV side. Since there are only two 13kV circuits and two 4kV circuits near the site location, there is a reduced flexibility of interconnecting to multiple nearby circuits. The BESS would not have the operational flexibility to connect to other distribution feeders without major street construction work. River street location would create an extended timeline for scope of work and development changes along with some financial liability constraints. Since the location is two miles away from the station, it does create additional electrical and system studies beyond the duct in manhole and construction work.

Furthermore, the BESS analyzed is not a multi-functional battery storage. At this point, it is purely for offloading during peak times. There were no micro grid, micro grid controlling, protections, or control scheming analyzed.

The River Street site can be assessed separately to address the request made by the City of Boston to reduce the electric outage frequency risk at this site. Following the River St property proposal, additional properties were recommended by the City of Boston for consideration. These properties, along with Eversource owned properties in the Hyde Park area, were reviewed and ranked in a Site Selection Matrix.

Traditional Solution Alternatives

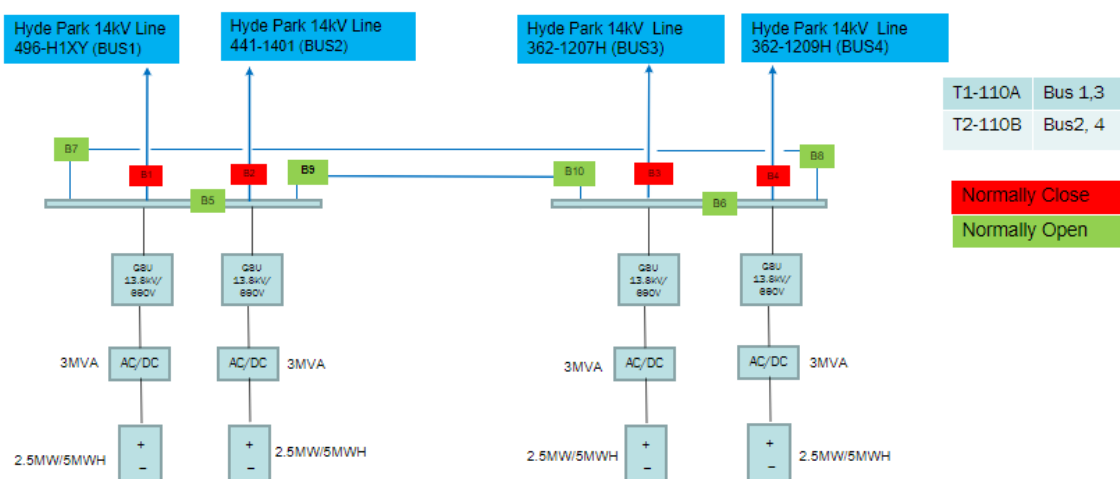
Alternative 1:

The proposed BESS system is expected to perform peak load shave functionality to relieve the Substation heavy loading condition.

Proposed BESS interconnection one line diagram:
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Hyde Park Proposed BESS One-Line Diagram 10MW/20MWh

EVERSOURCE
ENERGY



[END CEII]

Figure 15 Proposed BESS Interconnection One-Line

Power/ Energy Rating	Nameplate Real Power (P)	10 MW
	Nameplate Reactive Power(Q)	6 MVAR
	Nameplate Apparent Power(S)	12 MVA
	Nameplate Energy	20 MWh
	Quadrants of Operation (PCS)	4
DC Voltage	Voltage Range	TBD
AC Interconnection	Interconnection Voltage	13.8kV
	Nominal Frequency	60HZ
Ramp Rates	Operational Ramp Rate (Charging)	TBD
	Operational Ramp Rate (Discharging)	TBD
Battery Cycling	Primary use case	Peak Load Shave
Grid Support Functions	Black Start Capable	N/A
	Islanded Operation (Grid Forming V, F source)	N/A

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EVERSOURCE

Project Authorization Form

	Voltage Support /VAR Compensation	Yes
	Frequency Support	Yes
	Frequency Regulation	Yes
	Voltage ride through	Yes
	Frequency Ride through	Yes
	Inertia Emulation	Option Adder
	Adjustable V, F Windows	Option Adder
Environmental Conditions	Ambient Temperature	-40C to + 40C (50 C as an option adder)
	Relative Humidity	100%
	Altitude	200' above sea level
	Coastal, Salt Air, Ocean Fog	No
	Wind Load	150 mph
	Design Snow Load	25 psf
	Seismic zone	C
	Sound requirement	Not to exceed 64 dB (A) at operating mode that results in maximum sound output <u>at</u> the fence line of the system.
	Building Footprint	N/A
Performance, Warranty, Availability	AC Round trip efficiency	>85%
	System Lifetime	20 years
	Warranty	10 years
	Availability	>99%
	Capacity Assurance	Yes

Table 12 Hyde Park BESS Proposed Technical Specifications

Conceptual (-25%/+50%) BESS Cost Estimate: \$44.379 million
 Conceptual (-25%/+50%) Distribution Scope Cost Estimate: \$4.793 million
 Conceptual (-25%/+50%) Total BESS Cost Estimate: \$49.2 million total
 The scope of work and cost for the distribution work to interconnect the BESS will be submitted and approved through MPAC and is included in the above Alternative 1 cost.

21334-Hyde Park BESS IFR_R4 and Hyde Park Steady State Report are attached.

Advantages :

- BESS will help relieve heavily loaded Hyde Park Substation.
- Construction is confined to a single site where the batteries can each interconnect to four different 13.8kV Hyde Park distribution feeders and provide substation bus diversity.
- Can be completed in a minimum of 3 years.

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- This option has the least overall construction impact on the communities.
- Shorter timeline for Outreach as compared to other alternatives since discussions with the city of Boston is already underway. Expected timeline for outreach, which has already commenced is approximately 12 – 18 months before a filing, which would likely be around Q2/Q3 of 2024 and continue ongoing thereafter. As part of the overall Hyde Park Area Solution, which includes the BESS, outreach and community engagement has already begun with active engagement and participation from the City of Boston and Hyde Park neighborhoods. There has been early involvement with the city for this alternative and the site selection, which helps to smooth the path for siting the BESS with city support and reduces opposition.
- After the five year window between the BESS installation and the long-term project or new substation installation, the BESS can be redeployed to other locations where it can be useful.

Disadvantages:

- This alternative will require a City of Boston Article 80E Small Project Review and a variance with the Boston Zoning Board of Appeals. It is not known if previous utility only projects have been approved through this process but a review of previous projects by the Siting team has indicated an average timeframe for approval of 18 months.
- If the Article 80E and Zoning Relief process is not successful, a 40A zoning exemption would be required from the DPU which would add significant time to the project.
- There has been local and neighborhood opposition for a proposed housing development on an adjacent parcel and similar opposition and concerns are expected for the BESS development. Project Services and Community Relations has been engaging with the City on this issue and will be performing targeted community outreach for this project and the Hyde Park solutions as a whole.

Alternative 2:

Bring additional capacity to Hyde Park area by constructing two 13.8kV distribution lines from Needham Substation.

Conceptual (-25%/+50%) Cost Estimate: \$46.473 million total

Advantages:

- This alternative would relieve the heavily loaded Hyde Park Substation.
- A filing with the Energy Facilities Siting Board (EFSB) or DPU is not anticipated, since this is a distribution related solution.

Disadvantages:

- Construction will be crossing multiple towns, busy highways and bridges, having an adverse impact on the community.

This alternative will either take the same amount of time as alternative 1 or more, due to the extensive in-street work required for this Alternative. Project Services and Community Relations would have to undertake substantial community outreach and

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dialogue with City officials, which has not commenced. Expected timeline for outreach is approximately 18 – 24 months and continue ongoing thereafter. Extensive permitting would also be needed in order to cross multiple towns.

- Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected.
- Routing options would need to be discussed with cities and other stakeholders.
- Rights would need to be obtained from the towns/cities to install the conduits. Grant of location would be needed for Needham, Dedham and Boston.
- Permits would be needed to cross Route 128 and the bridges, and other entities such as MWRA or DCR. This can even take up to a year.
- The following caveats pose as disadvantages as well and can affect the cost estimates for this alternative:
 - South St in Needham has been recently paved, so it is uncertain whether curb to curb paving would be required. Costs to mill and pave this section are included in the estimate.
 - Two overhead lines will be installed to cross Route 128. This may result in increased labor costs and also decreases reliability of these circuits.
 - Conduit/pipes need to be installed over a bridge crossing at the Dedham/Needham line.
- Expect significant mitigation requests for roadwork restoration, such as curb-to-curb/gutter-to-gutter paving, ADA ramp restoration, new sidewalks, and/or traffic calming measures like speed humps, and other community benefits, which could include monetary payments for trees or other types of added green space.
- Depending on the extent of work required at the Needham and Hyde Park substations (not yet known), zoning relief may be required from the Town of Needham and the City of Boston.
- Based on the current forecast, this alternative would not remain as useful after the five year window between the construction of this alternative and the long-term project or new 115kV/13.8kV substation construction.

New Needham 13.8kV Feeder One Lines are attached.

Alternative Summary

Although the cost for the alternative 2 is lower than the preferred alternative, it is critical to note the hurdles and timing delay that would be faced if this alternative were to be implemented as explained above. Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected. There are also potential hurdles involved in gaining rights and permits for alternative 2. Alternative 2 may take significantly longer than the BESS to implement as explained above. Outreach for the BESS alternative 1 has already been ongoing which helps to smooth the path for siting the BESS with city support and reduces opposition whilst there has been no outreach for alternative 2.

Two of the feeders that the BESS would connect to would also benefit from the BESS, Circuit 496-H1XY supplied from Bus 1 historically runs 80-90% of its summer normal ampacity in the summer. The circuit 362-1209 supplied from Bus 4 historically runs 60-90% of its summer normal in the summer. The BESS would help to reduce the loading on these circuits.

The BESS can be redeployed to other stations where it can be useful after the five year window between the BESS installation and the long-term project construction. However, based on the current forecast, alternative 2 would not be as useful after the five year window when the long-term project or new 115/13.8 kV substation is in service to relieve load from Hyde Park and surrounding substations.

The River Street BESS solution was rejected as it did not include as much flexibility in connecting to different distribution circuits and would entail further System Planning studies and construction work, which would delay the project significantly. Such a delay would defeat the purpose of having a short term solution to the capacity issue at this substation. Both alternatives would involve supplying a minority Environmental Justice zone.

[4] Project Scope (Preferred Solution is Alternative 1)

- The Hyde Park BESS project consists of a 10MW/20MWh energy storage system. The Project will be located at 427 Cummins Highway in Boston on Eversource owned property. The Hyde Park BESS will be operated in parallel with the Owner's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8kV distribution infrastructure adjacent to the BESS facility.
- The BESS facility is proposed to interconnect to the following 13.8kV circuits supplied from Hyde Park substation:
 - 496-H1XY
 - 441-1401H
 - 362-1207H
 - 362-1209H
- Required Distribution System improvements include:
 - Extend distribution circuits 5,465' up to and into the Cummins Highway Property for the BESS interconnection. Scope of work includes three manholes, conduit, and associated cable.
 -
- BESS Preliminary Scope of Distribution Work
 - Manhole Installation (3) concrete manholes with inside dimension of 6'x10'x8'
 - Conduit Installation (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes
 - Primary Underground Cable Removal(13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)
 - Primary Underground Cable Installation(16,395') of 3-700 FS 15kV primary cable (47 sections)

Contains Critical Energy Infrastructure Information (CEII) - Do Not Release

- Interconnection to Circuit 496-H1XY
 - Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS (9 sections) and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
 - Note: 1 Single three-way joint in MH18478 and 3 single multiple joints in MH12361.
- Interconnection to Circuit 441-1401H
 - Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Interconnection to Circuit 362-1207H
 - Cut and remove cable from MH15729 to MH12361 (5 sections).
 - Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Interconnection to Circuit 362-1209H
 - Cut and remove cable from MH2434 to MH12358 (14 sections).
 - Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

[5] Project Schedule

Milestone	Scheduled Completion Date
IFR Approval	June 2021
SDC Approval	February 2024
Conceptual Design Completion	June 2025
EPAC Partial Funding Approval (for applicable projects)	May 2024
EPAC Full Funding Approval	August 2025
Start of Construction	June 2026
Project In-Service Date (ISD) ¹ Based on the current available information, Eversource can get materials and perform the work by the scheduled in-service date of September 2027. Once an EPC vendor is on board, this window will be reevaluated accordingly.	September 2027
Project Need Date (for applicable projects)	Present

[6] References

References
1. Section 2.3, DSPG 2020
2. Sections 2.3 and 2.7.1, DSPG 2020
3. Section 3.1 [of the DSPG]
4. Section 2.10, DSPG 2020
5. 21334-Hyde Park BESS IFR_R4

[7] Attachments (One-Line Diagrams, Images, etc.)

Attachments
A. Cost Estimates
B. Hyde Park Steady State Report
C. New Needham 13.8kV Feeder One Lines

[8] Preliminary Project Checklist (For the preferred solution)

PLANNING	
Is a NX-9 required?	No
Is an ISO-NE PAC presentation required?	No
Is a PPA required?	Yes
Is a TCA Application Required?	No
ENVIRONMENTAL JUSTICE COMMUNITIES	
Is any part of this project in an Environmental Justice Community?	Yes
Was the Equity Framework Checklist completed?	No
Has Project Services established an enhanced outreach plan per State Regulatory requirements?	Yes

Project Title: Battery Storage - Hyde Park

Project Mgr/Lead: Michael Rosenburg

Project Number: 21134

WO & Task #: 80363021

Estimate revision #: 00

Estimate Type - Station work

Estimate #: E-23-373

Date of Estimate: 12/08/2023

Estimate owner: PAR

In Service date: 10/23/2027

Template Revision: 6.01.00a

ESTIMATE SUMMARY

Estimate Type: Conceptual

WBS	Description	TOTAL	% of Directs	% of Total	Priors	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
1	ROW / Easements / Land purchases														
2	Environmental Approvals / Permits	\$310,000	1.1%	0.7%		\$14,200	\$74,700	\$102,600	\$118,500						
3	Outreach	\$276,200	0.9%	0.6%			\$58,600	\$63,000	\$123,900	\$30,700					
4	String Approvals / Permits	\$253,400	0.9%	0.6%			\$53,800	\$57,800	\$113,600	\$28,200					
5	Engineering / Design	\$790,800	2.7%	1.8%			\$348,300	\$360,200	\$17,400	\$34,900					
6	Materials (Eversource purchased)	\$13,761,800	46.6%	31.0%			\$4,451,800	\$4,678,400	\$4,631,600						
7	Construction (incl materials by contractor)	\$6,910,200	23.5%	15.6%			\$500	\$500	\$5,136,500	\$1,772,700					
8	Testing / Commissioning	\$1,063,800	3.6%	2.4%					\$311,500	\$752,300					
9	PMT Project Mgmt Team	\$1,049,400	3.6%	2.4%		\$11,200	\$55,900	\$133,600	\$468,000	\$380,700					
10	Removals	\$32,300	0.1%	0.1%	\$32,300										
11	Other (property taxes and BAR)	\$929,500	3.2%	2.1%		\$2,100	\$106,400	\$259,600	\$561,400						
12	Risks	\$4,000,000	13.6%	9.0%				\$950,000	\$1,650,000	\$1,400,000					
123	Project Directs Total w/Risks	\$29,377,400	100.0%	66.2%	\$32,300	\$27,500	\$5,150,000	\$6,635,700	\$13,132,400	\$4,399,500					
13	Indirects/Overhead	\$9,909,700		22.3%	\$49,500	\$61,200	\$942,000	\$1,133,300	\$5,343,100	\$2,380,600					
14	AFUDC	\$3,275,700		7.4%		\$8,800	\$218,600	\$729,900	\$1,660,900	\$657,500					
123	Project Total - Baseline Budget	\$42,562,800		95.9%	\$81,800	\$97,500	\$6,310,600	\$8,498,900	\$20,136,400	\$7,437,600					
15	Contingency	\$1,815,800		4.1%		\$1,500	\$461,500	\$482,000	\$751,900	\$119,900					
123	Project Total - Funding Approval Level	\$44,378,600		100.0%	\$81,800	\$99,000	\$6,772,100	\$8,980,900	\$20,888,300	\$7,556,500					
16	Reimbursables/Customer Contribution														
123	Project Total (less reimbursables)	\$44,378,600			\$81,800	\$99,000	\$6,772,100	\$8,980,900	\$20,888,300	\$7,556,500					
	Cost per key qty (if appl)	44,378,600	\$'s / LS BESS												
	Amount of Escalation included above	\$3,011,700					\$207,800	\$405,700	\$1,662,100	\$736,100					
6	Total materials - Eversource	\$13,761,800					\$4,451,800	\$4,678,400	\$4,631,600						
	Total materials - Contractor	\$33,500					\$500	\$500	\$24,300	\$8,200					
	Total materials	\$13,795,300					\$4,452,300	\$4,678,900	\$4,655,900	\$8,200					
	NOTE: Risk / Contingency is 15.1% of Project Total Costs (incl escal/indirects/AFUDC but excl reimb.)														
Estimate Range															
		-25%			50%										
		\$33,284,000			\$66,568,000										
COMMENTS:															
Project Name: Battery Energy Storage System															
Key qty's and UOM (tie to inputs on est sheets)						1	LS	BESS	3.22	Factor (total \$/Eversource mat'l \$'s)					
									2.13	Factor (Total directs \$/Total Eversource mat'l \$'s)					
15 MW/20MWh Battery Energy Storage System (BESS) - Cummings Hwy, Hyde Park SS #496, Boston MA.															
Installation of self contained battery enclosure															
Installation of Power Conversion system (PCS) units															
Installation of Step up transformers															
Installation of MV Switchgear															
Installation of underground ductbank to Manhole connection to grid.															
Installation of sound mitigation															
Installation of fire suppression systems															
Installation of grounding transformer and grounding grid															
Site development to include fencing and lighting															
Assumptions/Clarifications:															
Indirect rates based on: 210 - FMA Station work															
ISO is 4/23/2027 based on schedule provided by PMT.															
Prior costs are included as of 12/1/2023.															
Location is Eversource property. Footprint of BESS property will not impede future projects on this site.															
Labor and material was taken from company #3 out of the 5 on the BESS indicative pricing sheet, dated 7/28/2023, provided by PMT.															
Spent generation in two locations for load loss during construction.															
Construction to be external vendor.															
Engineering to be external with internal oversight.															
Initial testing and commissioning will be by external vendor.															
Initial training by external vendor															
Maintenance and upkeep of BESS is not included.															
RISKS															
Potential UG Obstructions			\$	350,000											
Severe Weather Delays NY			\$	600,000											
Design development			\$	500,000											
Additional outreach			\$	750,000											
Additional Siltation costs			\$	550,000											
Additional Environmental costs			\$	450,000											
Contaminated soil removal			\$	500,000											

All Estimate Summary for PP4 Calc (escalation separate)	
Estimate #: E-23-373	
Date of Estimate: 12/08/2023	
Project Number: 21134	
WO & Task #: 80363021	
TOTAL w/ escal broken out	
ROW / Easements / Land purchase	\$ -
Environmental Approvals / Permits	\$ 284,700
Outreach	\$ 250,300
String Approvals / Permits	\$ 229,400
Engineering / Design	\$ 734,600
Materials (Eversource purchased	\$ 13,098,800
Construction (incl materials by c	\$ 6,096,500
Testing / Commissioning	\$ 926,400
PMT Project Mgmt Team	\$ 931,400
Removals	\$ 32,300
Other (property taxes and BAR)	\$ 840,200
Risks	\$ 4,000,000
	\$ 27,424,600
Indirects/Overhead	\$ 8,851,100
AFUDC	\$ 3,275,700
	\$ 39,551,400
	\$ -
Contingency	\$ 1,815,800
	\$ 41,367,200
	\$ -
	\$ -
	\$ 3,011,700
Escal total	\$ 44,378,900
Check total	\$ 44,378,900

Project Number: 21334
Date of Estimate: 12/08/2023
Estimate revision #: 00

Indirect/Overhead Calculation Template

Line Work					Station Work						
	Year 1	Year 2	Year 3	Total		Year 1	Year 2	Year 3	Total	Project Total	
Labor	\$ 467,663			\$ 467,663	Labor	\$ -			\$ -	\$ 467,663	
Materials	\$ 928,459			\$ 928,459	Materials	\$ -			\$ -	\$ 928,459	
Outside Services	\$ 947,000			\$ 947,000	Outside Services	\$ -			\$ -	\$ 947,000	
Other (Salvage Credit)	\$ -			\$ -	Other	\$ -			\$ -	\$ -	
Total Direct Costs	\$ 2,343,122	\$ -	\$ -	\$ 2,343,122	Total Direct Costs	\$ -	\$ -	\$ -	\$ -	\$ 2,343,122	
Cust Contribution (negative number)	\$ -			\$ -	Customer Contribution				\$ -	\$ -	
NPT	\$ 86,050	\$ -	\$ -	\$ 86,050	NPT	\$ -	\$ -	\$ -	\$ -	\$ 86,050	
Payroll Benefits	\$ 109,635.18	\$ -	\$ -	\$ 109,635	Payroll Benefits	\$ -	\$ -	\$ -	\$ -	\$ 109,635	
MDEC	\$ 96,621	\$ -	\$ -	\$ 96,621	MDEC	\$ -	\$ -	\$ -	\$ -	\$ 96,621	
E&S	\$ 1,207,761	\$ -	\$ -	\$ 1,207,761	E&S	\$ -	\$ -	\$ -	\$ -	\$ 1,207,761	
AS&E	\$ 21,200	\$ -	\$ -	\$ 21,200	AS&E	\$ -	\$ -	\$ -	\$ -	\$ 21,200	
Lobby Stock	\$ 176,407.27	\$ -	\$ -	\$ 176,407	Lobby Stock	\$ -	\$ -	\$ -	\$ -	\$ 176,407	
Stores	\$ 111,415.12	\$ -	\$ -	\$ 111,415	Stores	\$ -	\$ -	\$ -	\$ -	\$ 111,415	
NSTAR Vehicles	\$ 99,144.56	\$ -	\$ -	\$ 99,145	NSTAR Vehicles	\$ -	\$ -	\$ -	\$ -	\$ 99,145	
Total Indirect Costs	\$ 1,908,234	\$ -	\$ -	\$ 1,908,234	Total Indirect Costs	\$ -	\$ -	\$ -	\$ -	\$ 1,908,234	
AFUDC	\$ 281,015	\$ -	\$ -	\$ 281,015	AFUDC	\$ -	\$ -	\$ -	\$ -	\$ 281,015	

Project Costs Summary				
Capital Costs	Year 2022	Year 2023	Year 2024	Total
Direct Capital Costs				
Internal Labor and Construction Equipment	\$ 467,663	\$ -	\$ -	\$ 467,663
Outside Services	\$ 947,000	\$ -	\$ -	\$ 947,000
Materials	\$ 928,459	\$ -	\$ -	\$ 928,459
Removals	\$ 261,300	\$ -	\$ -	\$ 261,300
Risk and Contingency		\$ -	\$ -	\$ -
Other	\$ -	\$ -	\$ -	\$ -
Subtotal	\$ 2,604,422	\$ -	\$ -	\$ 2,604,422
Indirect Capital Costs				
Indirect/Overhead (including benefits)	\$ 1,908,234	\$ -	\$ -	\$ 1,908,234
Capitlized interest or AFDUC, if any	\$ 281,015	\$ -	\$ -	\$ 281,015
Subtotal	\$ 2,189,249	\$ -	\$ -	\$ 2,189,249
Total Capital Costs	\$ 4,793,671	\$ -	\$ -	\$ 4,793,671
Total Customer Contribution	\$ -	\$ -	\$ -	\$ -
Total Capital Project Costs	\$ 4,793,671	\$ -	\$ -	\$ 4,793,671

**PAF required if Total Capital Costs exceed \$500k

Indirect/Overhead Calculation Template

Line Work				
	Year 1	Year 2	Year 3	Total
Labor	\$ 587,739			\$ 587,739
Materials	\$ 2,791,596			\$ 2,791,596
Outside Services	\$ 21,055,100			\$ 21,055,100
Other (Salvage Credit)	\$ -			\$ -
Total Direct Costs	\$ 24,434,435	\$ -	\$ -	\$ 24,434,435
Cust Contribution (negative number)	\$ -			\$ -
NPT	\$ 108,144	\$ -	\$ -	\$ 108,144
Payroll Benefits	\$ 137,784.74	\$ -	\$ -	\$ 137,785
MDEC	\$ 1,313,326	\$ -	\$ -	\$ 1,313,326
E&S	\$ 16,416,575	\$ -	\$ -	\$ 16,416,575
AS&E	\$ 191,593	\$ -	\$ -	\$ 191,593
Lobby Stock	\$ 530,403.21	\$ -	\$ -	\$ 530,403
Stores	\$ 334,991.50	\$ -	\$ -	\$ 334,992
NSTAR Vehicles	\$ 124,600.59	\$ -	\$ -	\$ 124,601
Total Indirect Costs	\$ 19,157,419	\$ -	\$ -	\$ 19,157,419
AFUDC	\$ 2,881,421	\$ -	\$ -	\$ 2,881,421

Project Costs Summary				
Capital Costs	Year 2022	Year 2023	Year 2024	Total
Direct Capital Costs				
Internal Labor and Construction Equipment	\$ 587,739	\$ -	\$ -	\$ 587,739
Outside Services	\$ 21,055,100	\$ -	\$ -	\$ 21,055,100
Materials	\$ 2,791,596	\$ -	\$ -	\$ 2,791,596
Removals	\$ -	\$ -	\$ -	\$ -
Risk and Contingency	\$ -	\$ -	\$ -	\$ -
Other	\$ -	\$ -	\$ -	\$ -
Subtotal	\$ 24,434,435	\$ -	\$ -	\$ 24,434,435
Indirect Capital Costs				
Indirect/Overhead (including benefits)	\$ 19,157,419	\$ -	\$ -	\$ 19,157,419
Capitlized interest or AFDUC, if any	\$ 2,881,421	\$ -	\$ -	\$ 2,881,421
Subtotal	\$ 22,038,840	\$ -	\$ -	\$ 22,038,840
Total Capital Costs	\$ 46,473,275	\$ -	\$ -	\$ 46,473,275
Total Customer Contribution	\$ -	\$ -	\$ -	\$ -
Total Capital Project Costs	\$ 46,473,275	\$ -	\$ -	\$ 46,473,275

Station Work					Project Total
	Year 1	Year 2	Year 3	Total	
Labor	\$ -			\$ -	\$ 587,739
Materials	\$ -			\$ -	\$ 2,791,596
Outside Services	\$ -			\$ -	\$ 21,055,100
Other	\$ -			\$ -	\$ -
Total Direct Costs	\$ -	\$ -	\$ -	\$ -	\$ 24,434,435
Customer Contribution				\$ -	\$ -
NPT	\$ -	\$ -	\$ -	\$ -	\$ 108,144
Payroll Benefits	\$ -	\$ -	\$ -	\$ -	\$ 137,785
MDEC	\$ -	\$ -	\$ -	\$ -	\$ 1,313,326
E&S	\$ -	\$ -	\$ -	\$ -	\$ 16,416,575
AS&E	\$ -	\$ -	\$ -	\$ -	\$ 191,593
Lobby Stock	\$ -	\$ -	\$ -	\$ -	\$ 530,403
Stores	\$ -	\$ -	\$ -	\$ -	\$ 334,992
NSTAR Vehicles	\$ -	\$ -	\$ -	\$ -	\$ 124,601
Total Indirect Costs	\$ -	\$ -	\$ -	\$ -	\$ 19,157,419
AFUDC	\$ -	\$ -	\$ -	\$ -	\$ 2,881,421

**PAF required if Total Capital Costs exceed \$500k



Hyde Park Substation Battery Energy Storage System (BESS)

Distribution System Impact Study: Steady State and
Dynamic Analysis

Report

System Planning MA

Prepared By: Mariam Chatha and Souresh Mukherjee

Revision History

Date	Rev.	Description
	1	Report

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Executive Summary

This report includes the dynamic analysis performed for the Battery Energy Storage System (BESS) being interconnected at four 13.8 kV Hyde Park distribution feeders at Cummins Highway in Boston on Eversource owned property. The four feeders are 496-H1XY, 441-1401, 362-1209H and 362-1207H supplied from Hyde Park Substation. There will be a 2.5MW/5MWH battery on each of the four feeders making the project consist of a total of 10 MW/20MWH Battery Storage. The Hyde Park BESS will be operated in parallel with Eversource's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8 kV distribution infrastructure adjacent to the BESS facility.

The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the substation peak. The substation peak exceeds the firm capacity of the station and is forecasted to keep rising due to step loads and EV charging load. The Battery can ride through multi-day peak load events. BESS can be connected to Baker St#110 side in the future through 25kV or 4kV feeders to support Baker Substation if needed, which will require underground cable construction.

Below is the preliminary distribution scope of work for the BESS interconnection to the four feeders.

Manhole Installation

-(3) concrete manholes with inside dimension of 6'x10'x8'

Conduit Installation

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes Note: this second duct bank is required only if the substation is constructed

-(60') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

Primary Underground Cable Removal

-(13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

Primary Underground Cable Installation

-(16,395') of 3-700 FS 15kV primary cable (47 sections)

Below is the scope of work for connecting the BESS to feeder 496-H1XY.

- Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS (9 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

- Note: 1 Single three-way joint in MH18478 and 3 single multiple joints in MH12361.

Below is the scope of work for connecting the BESS to feeder 441-1401H.

- Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1207H.

- Cut and remove cable from MH15729 to MH12361 (5 sections).

- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1209H.

- Cut and remove cable from MH2434 to MH12358 (14 sections).

- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

To study the impact of the BESS on the Hyde Park distribution system, a transient dynamic analysis of the preferred alternative was performed in the PSCAD (Power Systems Computer Aided Design) tool. The four Hyde Park 13.8 kV distribution feeders along with the proposed BESS systems were modeled¹ in PSCAD by using circuit load and impedance data collected from steady-state Synergi models. The dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis were done to verify compliance with IEEE 1547-2018 Standard.

The ROI analysis determined the BESS inverter's response to unintended island conditions. According to the IEEE standard, for an unintended islanding condition, the distributed energy resource (which in this case is the BESS) shall detect the island, cease to energize the island, and trip within 2 seconds of the formation of the island.

¹ The BESS systems on the distribution circuits were modelled with a generic inverter model, not specific to any one manufacturer. After a vendor for the BESS is selected, the dynamic study will have to be re-run with the manufacturer specific inverter model.

When interconnecting distributed energy resource there is a concern that during step changes in load (such as tripping of an upstream device), the proposed inverters may cause transient over voltages in excess of 1.2 per unit, which can potentially cause damage to the customer's equipment and/or utility equipment. The TOV analysis was conducted to verify if the expected transient overvoltage conditions caused by the BESS inverters during step changes in load were within standard limits.

The ROI and the TOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios. The N-1 operational scenarios included one of the four feeder outages, a transformer outage, and a bus outage. The BESS system passed the ROI test for all scenarios tested. The BESS system also does not contribute to any TOV conditions for any of the tested scenarios.

Risk of Islanding (ROI) Screening

Based on the suggested guidelines for anti-islanding screening from Sandia National Laboratories and Eversource's distributed energy resources (DER) screening for ROI assessment (shown in Figure 1), additional requirements of ROI are necessary. Further study was performed to determine if the BESS inverters are reacting appropriately and not cause violations.

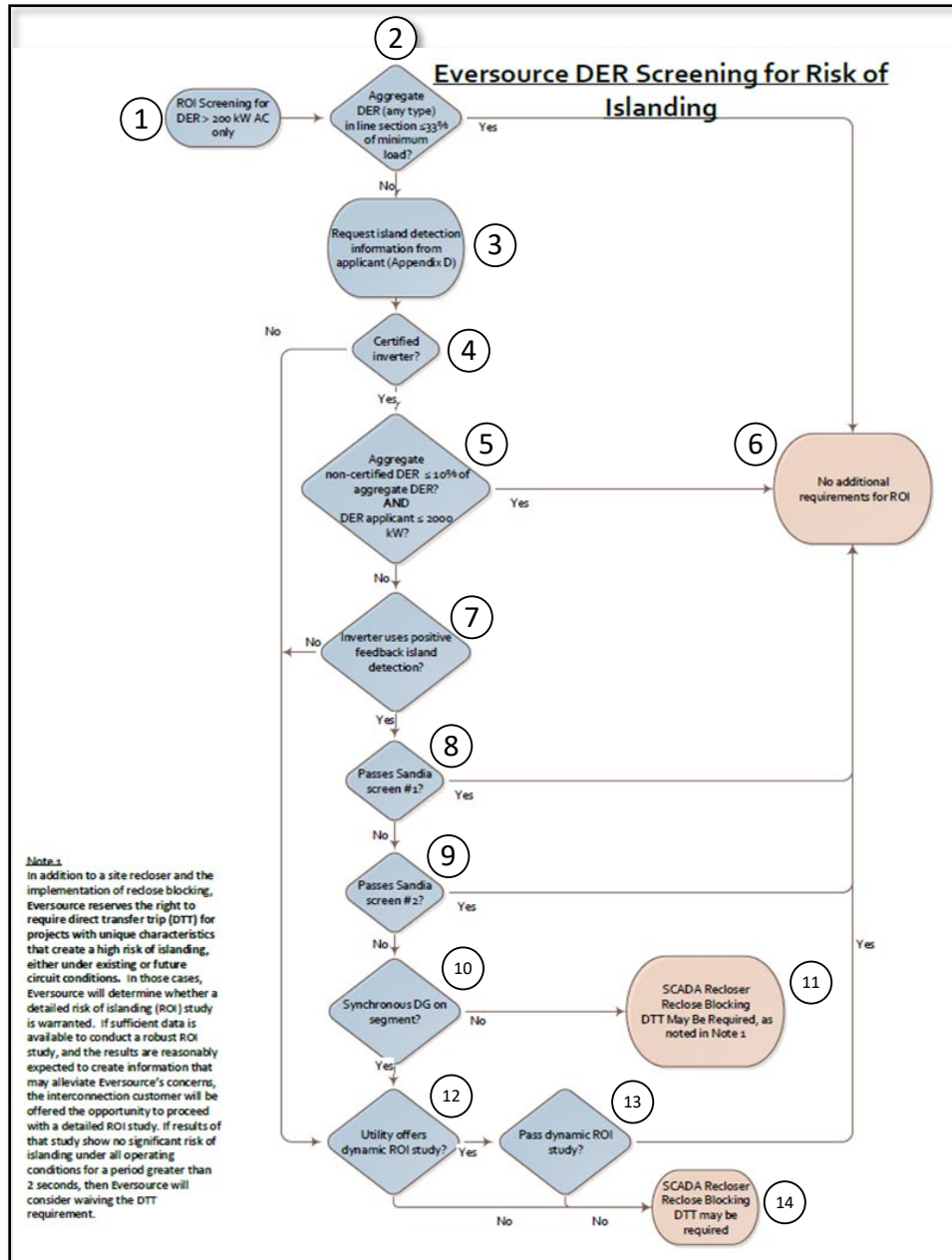


FIGURE 1. RISK OF ISLANDING (ROI) SCREENING PROCESS

During the dynamic study, in PSCAD, a detailed risk of islanding analysis was performed. The BESS system passed the ROI test for all four feeder interconnections.

Feeder	DER Interconnection	ROI Test
441-1401H	2.5 MW BESS	PASS
496-H1XY	2.5 MW BESS	PASS
362-1207H	2.5 MW BESS	PASS
362-1209H	2.5 MW BESS	PASS

TABLE 1: HYDE PARK BESS DYNAMIC STUDY – ROI RESULTS

PSCAD analysis shows that the BESS inverter trips within 2 seconds of unintentional island formation.

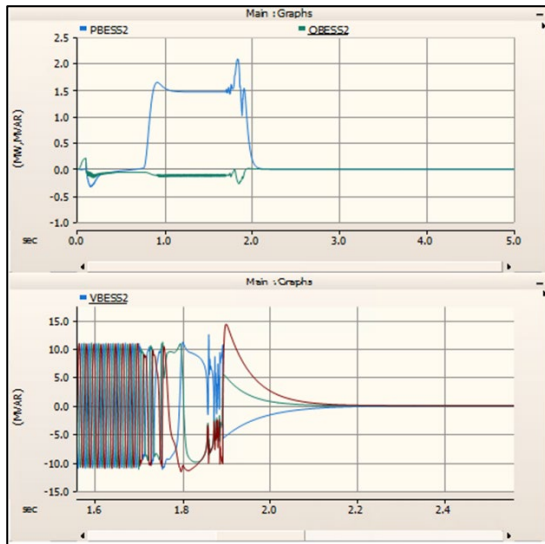


FIGURE 2: ROI TEST FOR FEEDER 441-1401H

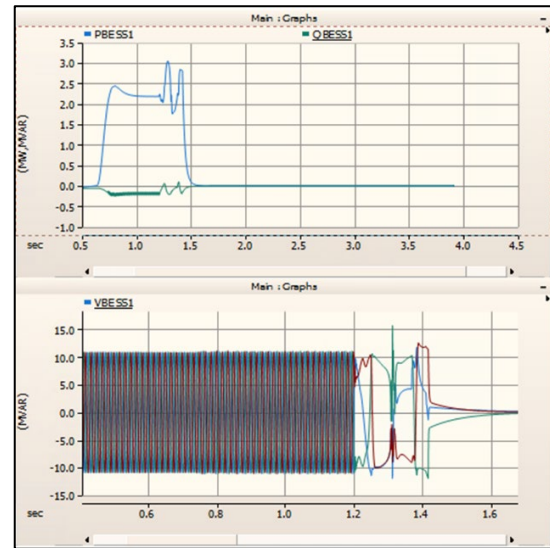


FIGURE 3: ROI TEST FOR FEEDER 496-H1XY

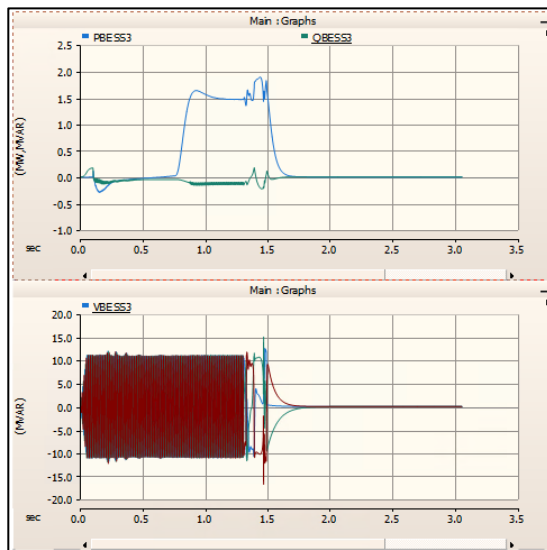


FIGURE 4: ROI TEST FOR FEEDER 362-1207H

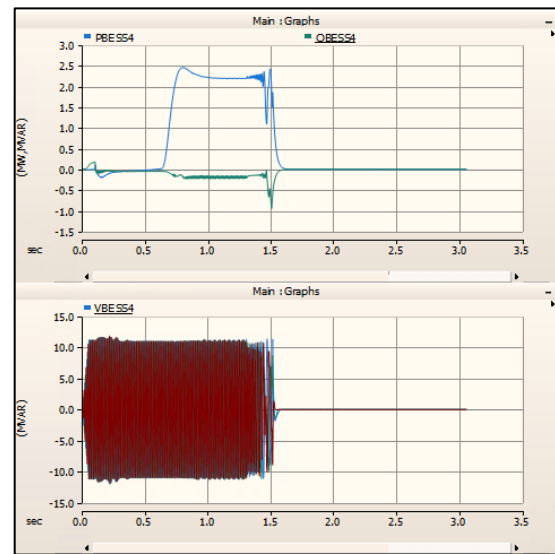


FIGURE 5: ROI TEST FOR FEEDER 362-1209H

Transient Overvoltage (TOV) Screening

If aggregate DER is greater than or equal to 115% of minimum gross load of the circuit, then there is a risk of Transient Overvoltage (TOV) at the feeder level and/or substation level. The same PSCAD dynamic study required for Risk of Islanding is further necessary to evaluate the TOV concerns. For dynamic transient overvoltage study, circuit load and impedance data are collected from steady-state Synergi models and used to build dynamic feeder models. The BESS inverters were modelled using generic PSCAD models. All other generators on the distribution circuits are modelled with a generic inverter model, not specific to any one manufacturer. The TOV analysis includes both Load Rejection Overvoltage (LROV) and Ground Fault Overvoltage (GFOV) studies.

Load Rejection Overvoltage (LROV) occurs when an upstream circuit breaker or recloser opens, which can cause an overvoltage on the isolated circuit due to stored energy in line and transformer inductances. LROV is modelled in PSCAD by operating all generation at 100% output in a minimum load case and opening the feeder circuit breaker. The voltage at the Point of Interconnection (POI) is measured.

Ground Fault Overvoltage (GFOV) is simulated by using the same minimum load, maximum generation scenario as the TOV scenario. A single-line-to-ground fault is initiated at the high side of the BESS interconnection transformer which persists until the upstream feeder breaker interrupts the fault after 10 cycles (0.16667 seconds). This is a conservative simulation and models a failure of protection selectivity.

Both LROV and GFOV durations must not exceed the following limits specified in IEEE 1547.

Transient Overvoltage Magnitude (per unit)	Maximum Allowable Duration (ms)
1.2	166
1.3	16
1.4	3
1.7	1.6
2.0	0

TABLE 2: IEEE 1547 STANDARD

Both, the LROV and GFOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios which included any one element out of service (OOS). The N-1 operational scenarios included:

- i. Transformer #1 OOS
- ii. Bus #1 OOS
- iii. Bus #3 OOS

The normal configuration of the proposed BESS at Hyde Park station is shown on the one-line diagram in figure 6. The N-1 contingency scenarios and corresponding breaker operations are shown in table 3.

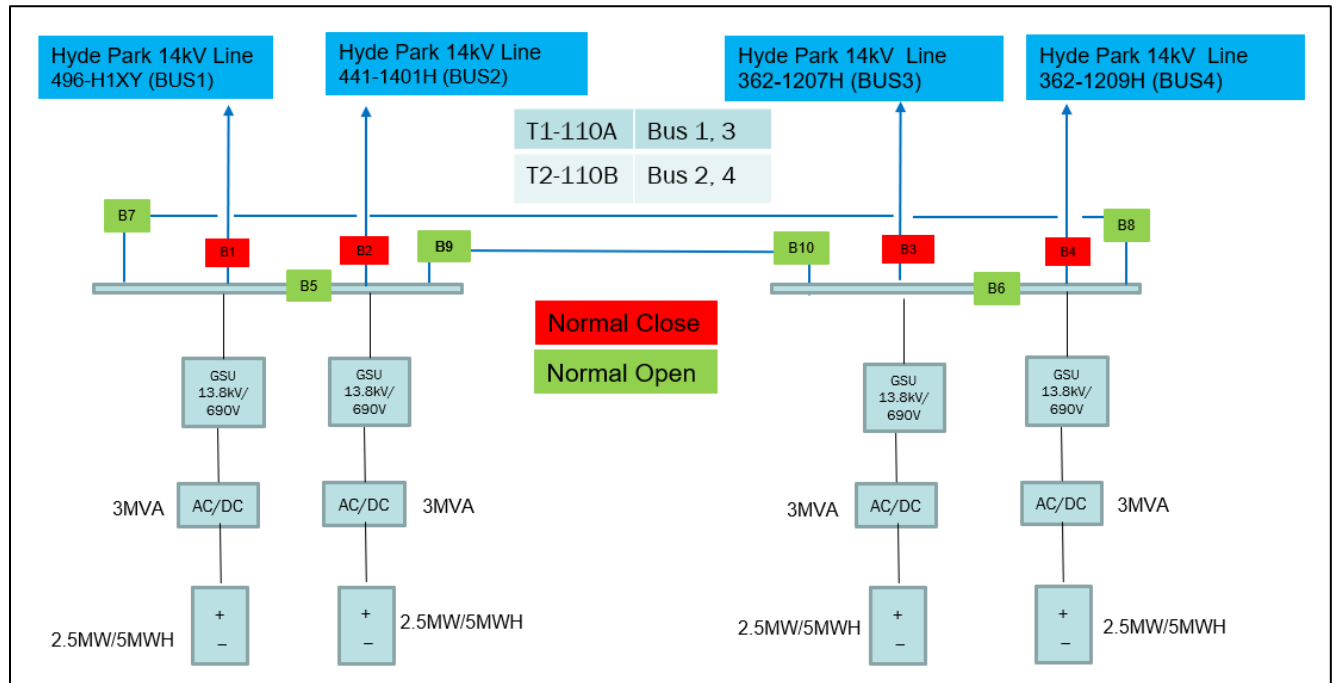


FIGURE 6: HYDE PARK PROPOSED BESS ONE-LINE DIAGRAM

Scenarios	Close	Open
Normal	B1, B2, B3, B4	B5, B6, B7, B8, B9, B10
T1- 110A OOS	B2, B4, B5, B6	B1, B3, B7, B8, B9, B10
Bus #1 OOS	B2, B3, B4, B5	B1, B6, B7, B8, B9, B10
Bus #3 OOS	B1, B2, B4, B6	B3, B5, B7, B8, B9, B10

TABLE 3: LIST OF TEST SCENARIOS

Results at Project (BESS) POI under Normal Configuration (N-0):

Load Rejection Overvoltage (LROV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 441-1401H.

Pre-project					Post-project				
Feeder 441-1401H					Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.56	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 441-1401H.

Pre-project					Post-project				
Feeder 441-1401H					Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 496-H1XY.

Pre-project					Post-project				
Feeder 496-H1XY					Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	2.10	1.80	0
1.3 pu	16	0	0	0	1.3 pu	16	1.66	1.36	0
1.4 pu	3	0	0	0	1.4 pu	3	1.26	0.98	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0.22	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 496-H1XY.

Pre-project					Post-project				
Feeder 496-H1XY					Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	24.54
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.52
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.20	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.02
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	36.76	0	0
1.3 pu	16	0	0	0	1.3 pu	16	1.42	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	25.84	0
1.3 pu	16	0	0	0	1.3 pu	16	0	7.88	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Results at Project (BESS) POI under several Contingency Configurations (N-1):

(i) Transformer #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.04	4.42	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(ii) Bus #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.08	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(iii) Bus #3 Out of Service:

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.02	4.20	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Steady State Analysis Summary Results

Power Flow Study

The below graphs show that the four (4) 2.5MW/5 MWh battery units can reduce the peak load to approximately 95% of the Load Carrying Capacity of the Hyde Park Substation in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 14kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders.

The project is designed to meet the 10 year forecast, where the highest load being is in 2032. Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any batteries installed. The batteries act as a load and charge from 04:00 **AM** to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load as shown below.

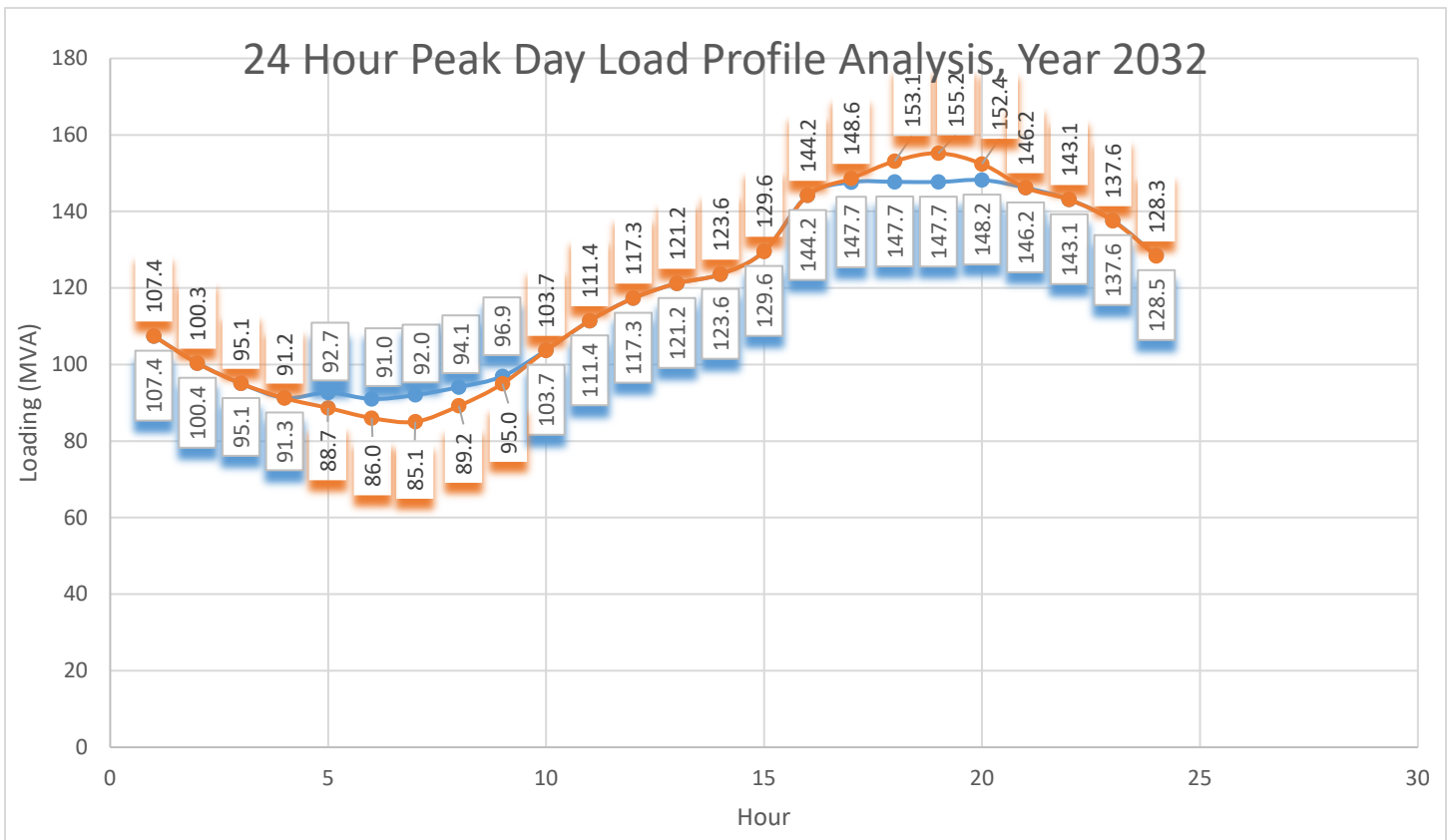


Figure 7

The orange line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decrease load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

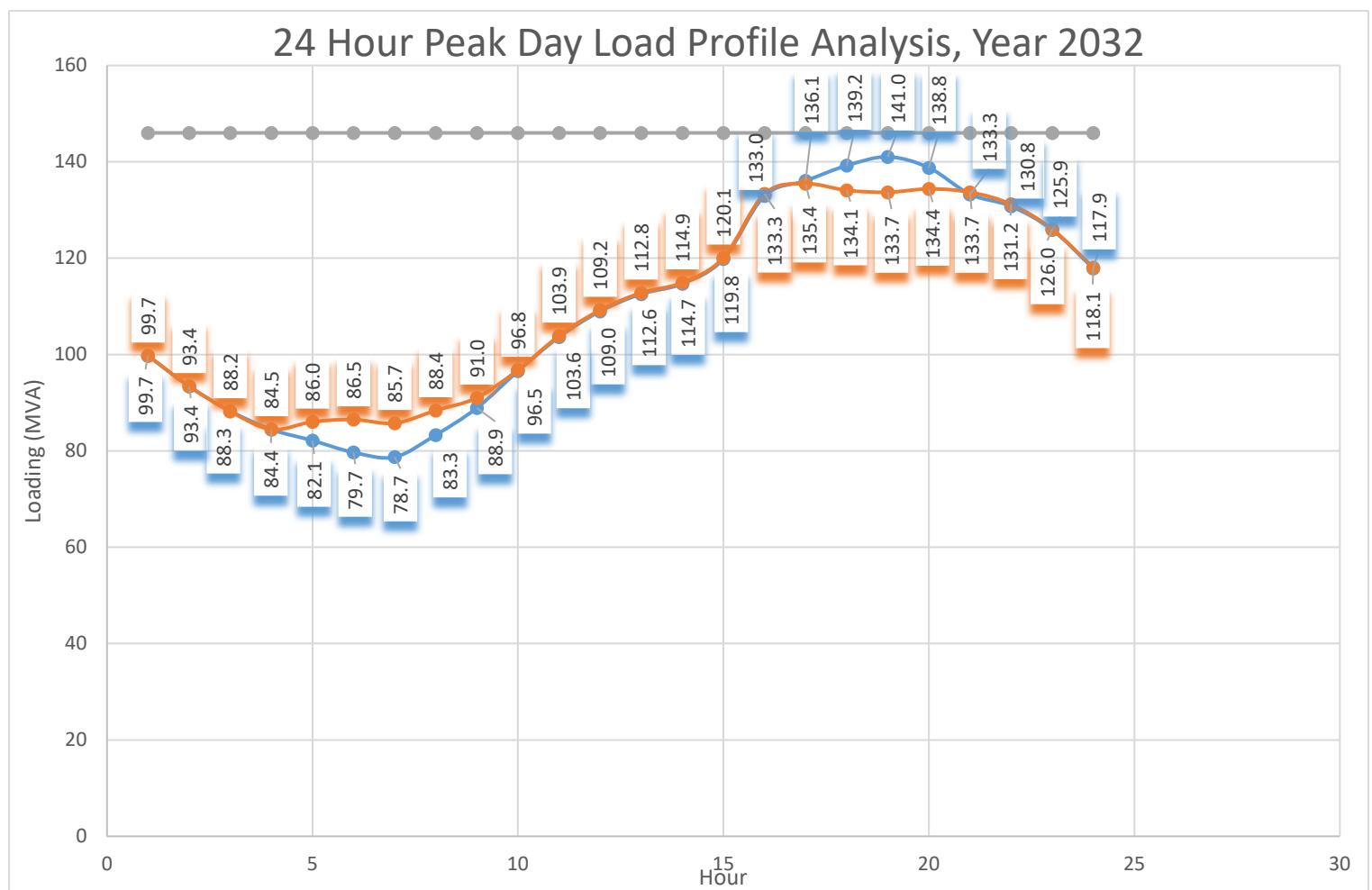


Figure 8

Thermal Rating Analysis

Below graph shows the 2032 24 hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational. It shows the 24 hour peak load

day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

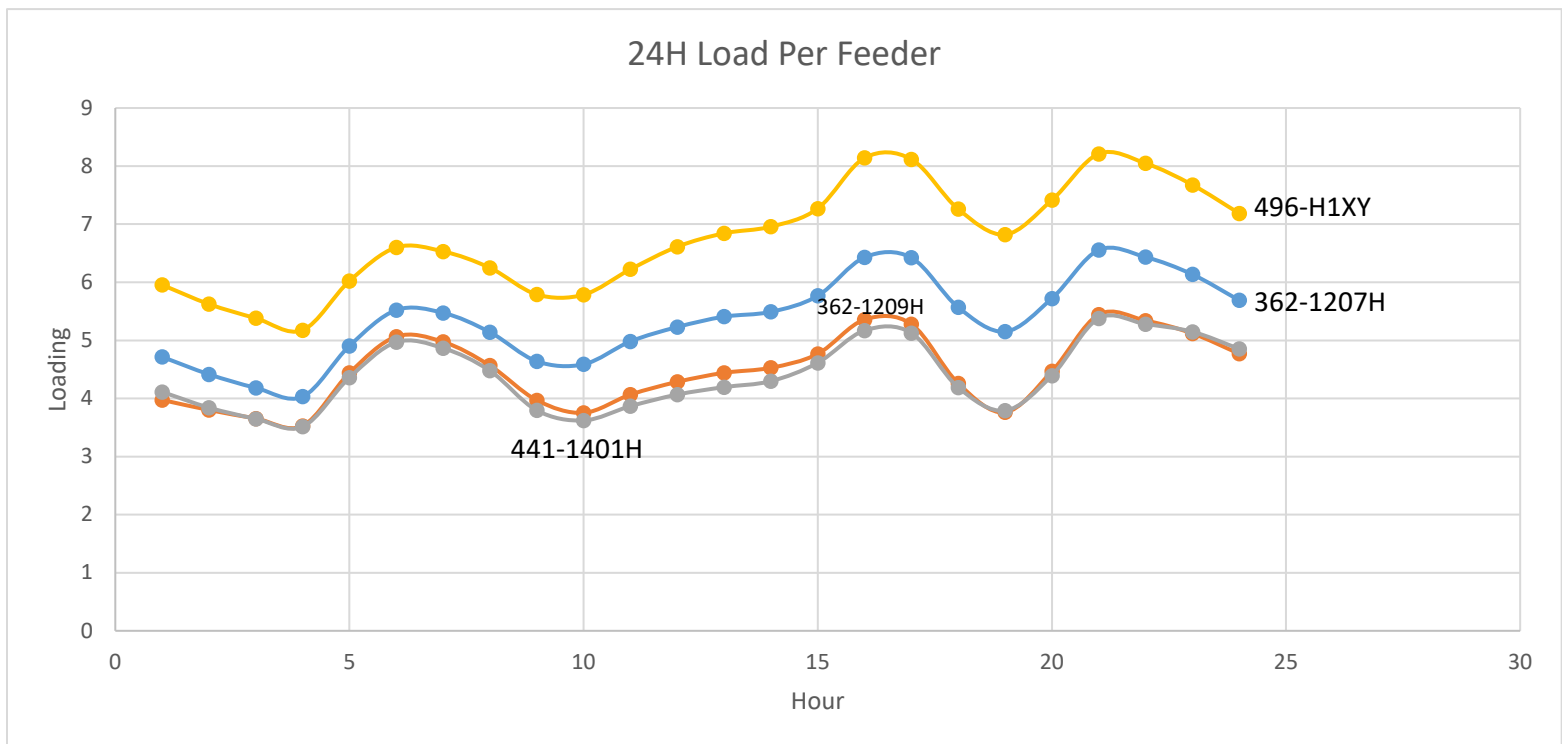


Figure 9

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.9	4.4	4.4	6.0
6:00 AM	5.5	5.1	5.0	6.6
7:00 AM	5.5	5.0	4.9	6.5
8:00 AM	5.1	4.6	4.5	6.3
9:00 AM	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The exact loading for the year 2032 cannot be predicted at this stage. However, based on the above loading data, there is a flexibility to charge the BESS on three of the above feeders instead of four if need be. Based on the above loading, feeder 441-1401 has the lowest amount of load and would be able to accommodate another 2.5 MW of BESS while feeder 496-H1XY may not. System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

Below is a graph showing circuit loading for each of the four feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four feeders. It shows the 24 hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

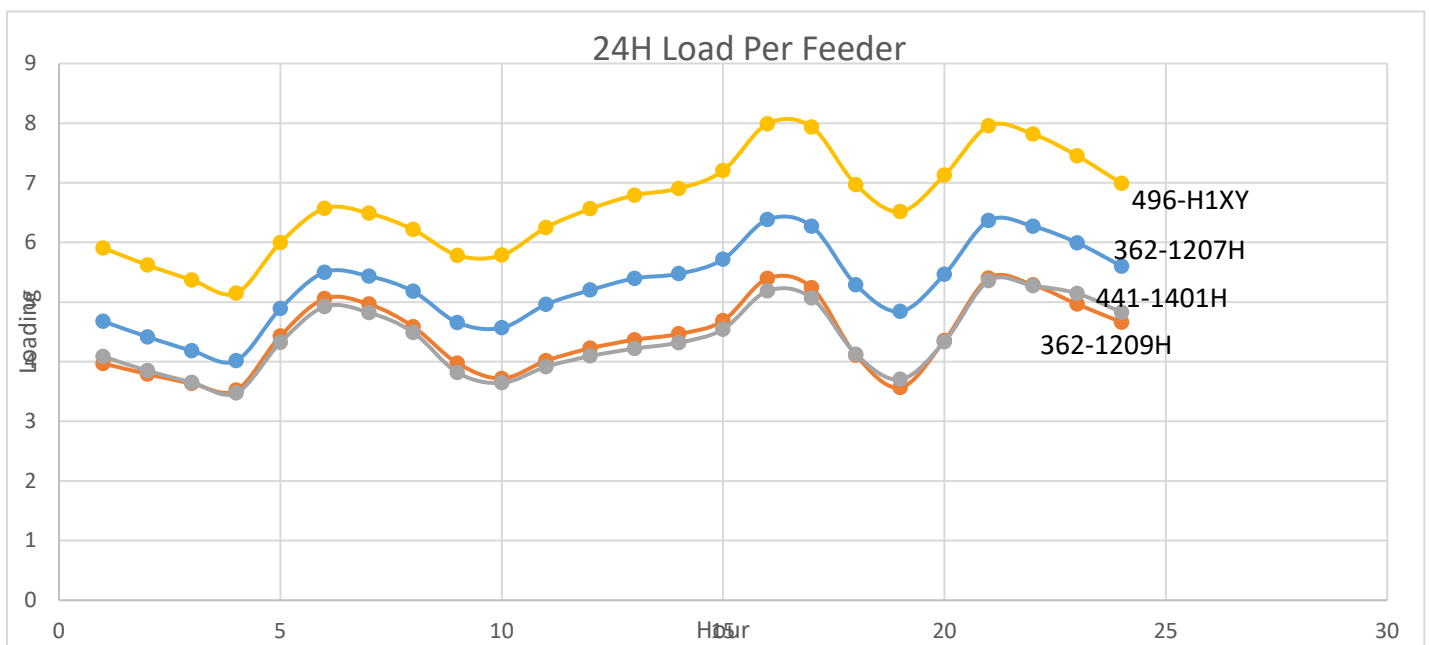


Figure 10

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.89	4.43	4.32	5.996

6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The BESS also helps to bring down the peak load on these 14 kV feeders. The Hyde Park circuit 496-H1XY historically runs 80-90% of its summer normal ampacity in the summer. The circuit 362-1207 supplied historically runs 60-80% of its summer normal in the summer months. The circuit 362-1209 historically runs 60-90% of its summer normal in the summer months.

Reverse Power Flow Evaluation

Due to the small size of the battery, there is no reverse power flow into the Hyde Park substation transformers. To evaluate the reverse power flow, the worst case scenario was analyzed where two batteries with a total size of 5MW/10MWH were connected to one feeder and the BESS was operating at a 100% output. This is assuming there is a contingency on one of the four 14 kV feeders and there would need to be two 2.5MW/5MWH BESS on one feeder. It is seen that the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery. Therefore, there is no reverse power flow because the battery would just decrease the load and is not large enough to cause reverse power flow into the electric grid.

For example, the below tables show the loading on the transformer supplying feeder 496-H1XY before and after 5MW/10MWH BESS was connected to it and operating at 100%output.

Transformer: STA_496_110A_2

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Transformer	View analysis results					
LTC / LDC	Result	A/AB	B/BC	C/CA	Avg	Mx/T
Settings	▼ General					
Tertiary	Ft Into	495578	465396	485737	482237	495578
Net Protector	Ft Away	---	---	---	---	0
Rates	Nom kV	ABC	ABCN	---	---	13.80
Info	▼ Taps and LTC					
Results	Vrr	---	121.7	---	121.7	---
	Amps Out	1599	1611	1615	1608	1615
	Tap Pos	11	11	11	11	11
	▼ Volts and Amps					
	Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
	Volts Out	121.8	121.7	121.7	121.8	121.8
	Amps Into	206	206	207	206	207
	Pct pf	92.0	92.0	92.0	92.0	---
	Imbalance	V,I,In	0.0%	0.3%	0 A	---
	▼ Power Flow					
	MVA Into	13.70	13.70	13.70	13.70	41.10
	MW Into	12.60	12.60	12.60	12.60	37.80
	Mvar Into	5.38	5.38	5.38	5.38	16.14
	MW Demand	12.46	12.55	12.58	12.53	37.59
	Mvar Demand	3.46	3.47	3.47	3.47	10.40
	Nom kvar Into	D,S,LC	0	0	0	---
	Nom kW Into	D,S,LC	0	0	0	---

Power Flow Without BESS

Transformer: STA_496_110A_2



Transformer	View analysis results
LTC / LDC	
Settings	
Tertiary	
Net Protector	
Rates	
Info	
Results	

Result	A/AB	B/BC	C/CA	Avg	Mx/T
General					
Ft Into	495578	465396	485737	482237	495578
Ft Away	---	---	---	---	0
Nom kV	ABC	ABCN	---	---	13.80
Taps and LTC					
Vrr	---	121.8	---	121.8	---
Amps Out	1412	1424	1427	1421	1427
Tap Pos	11	11	11	11	11
Volts and Amps					
Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
Volts Out	121.8	121.8	121.8	121.8	121.8
Amps Into	182	182	183	182	183
Pct pf	90.4	90.4	90.4	90.4	---
Imbalance	V,I,In	0.0%	0.3%	0 A	---
Power Flow					
MVA Into	12.11	12.11	12.11	12.11	36.32
MW Into	10.94	10.94	10.94	10.94	32.83
Mvar Into	5.18	5.18	5.18	5.18	15.55
MW Demand	10.81	10.91	10.94	10.89	32.66
Mvar Demand	3.69	3.69	3.69	3.69	11.07
Nom kvar Into	D,S,LC	0	0	0	---
Nom kW Into	D,S,LC	0	0	0	---

Power Flow With BESS

As shown, the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery and there is no reverse power flow.

Flicker

The flicker values comply with the Eversource's allowable limit of 2.0% at the distribution circuits. The voltage difference when the BESS is off and when the BESS is operating at 100% output was analyzed and was below 2% during normal and contingency scenarios. The worst case scenario where there are two BESS with a total size of 5 MW/10MWH operating at 100% output on one feeder was also analyzed. The voltage difference still remained below 2%. This voltage difference was analyzed at the peak load of 155 MVA.

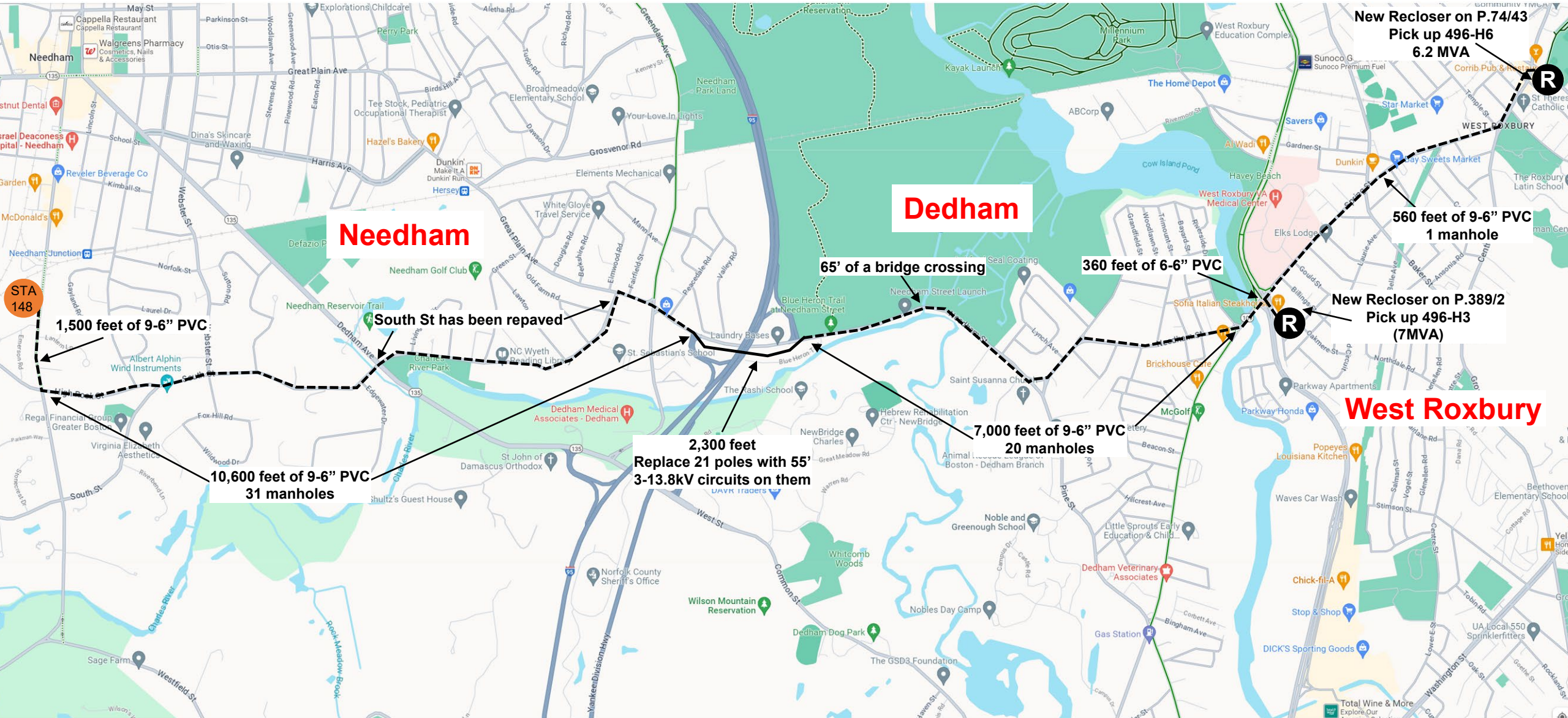
Transformer LTC Impact

The tap position range from -16 to 16 and the voltage is maintained at 122 Volts at the 14 kV buses. When the BESS is connected, the voltage increases slightly due to the decrease in load. Therefore, the BESS may cause the LTC to move to a slightly lower tap position to maintain the voltage at 122 volts. Overall, due to the small size of the battery, it would not have much impact on the transformer LTC tap positions.

Short Circuit

Due to the small size of the battery, it would not have much impact on the short circuit current.

Conduit and Route from STA 148 to 496-H3 & 496-H6



Scope of Work to relieve STA 496 via STA 148.

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit

21 (55 foot poles)

To Relieve 496-H3 (7 MVA)

19,900 feet of 3-700FS

2,450 feet of 477 ACSR

2 New Reclosers

2 New Underslung

2 new 45 foot poles

To Relieve 496-H6 (6.2 MVA)

25,900 feet of 3-700FS

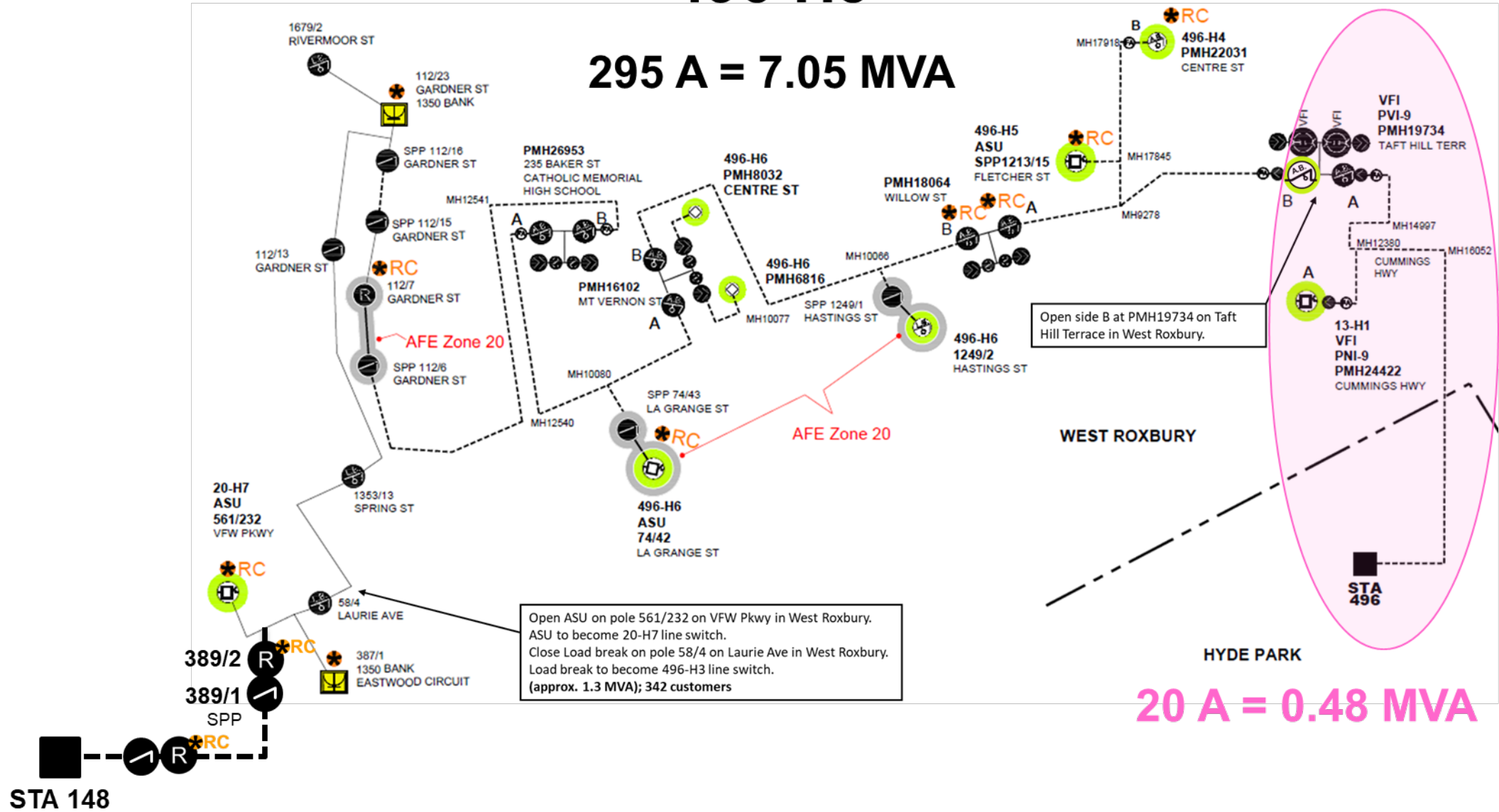
2,300 feet of 477 ACSR

2 New Reclosers

2 new Underslung

496-H3

295 A = 7.05 MVA



496-H6

260 A = 6.21 MVA

STA 148

20 A = 0.48 MVA

